# CARLOS EMR
# Release 2026.08 Full Verification Coverage Plan

Executable plan for an implementing agent: Java unit, Java integration (H2 and native MariaDB), Node and Playwright coverage proving that every change on `release/2026.08` not yet on `main` is correct, complete, free of unintended breaking changes and free of silent failures.

| | |
|---|---|
| **Scope** | `origin/main` e53e6cb5 (2026.08.0-beta6) .. `origin/release/2026.08` 710fb1ac (2026.08.0-beta7-SNAPSHOT) |
| **Delta** | 115 commits (110 first-parent), 1,924 files, +73,461 / −16,856 lines |
| **System of record** | `docs/testing/release-2026.08-verification.json` (40 workstreams, 52 checks) and `scripts/ci/verify-release-evidence.py` |
| **Source file** | `docs/superpowers/plans/2026-11-06-release-2026.08-full-verification-coverage.md` |
| **Prepared** | 2026-11-06, measured against the repository at the SHAs above; audited against the release tree the same day |

## Contents


- Release 2026.08 Full Verification Coverage Plan
  - 0. Orientation for the executing agent
  - 1. Scope inventory (measured 2026-11-06)
  - Global Constraints
  - Review Focus
  - Phase 0: Environment and baseline evidence
  - Phase 1: Traceability
  - Phase 2: Java unit coverage gaps
  - Phase 3: Java integration and native MariaDB
  - Phase 4: Node and Python
  - Phase 5: Playwright against the packaged install
  - Phase 6: Breaking-change and silent-failure sweeps
  - Phase 7: External checks, CI proposal, final verdict
  - Self-review notes
  - Appendix A: Commit to test-layer map (`origin/main..origin/release/2026.08`, first-parent)
  - Appendix B: Command reference


# Release 2026.08 Full Verification Coverage Plan

> **For agentic workers:** REQUIRED SUB-SKILL: Use superpowers:subagent-driven-development (recommended) or superpowers:executing-plans to implement this plan task-by-task. Steps use checkbox (`- [ ]`) syntax for tracking.

**Goal:** Produce executable, source-bound evidence that every change on `release/2026.08` that is not on `main` is correct, complete, free of unintended breaking changes and free of silent failures, across four layers: Java unit, Java integration (H2 and native MariaDB), Node, and Playwright against the packaged install.

**Architecture:** The release branch already carries the system of record for this work (PR #5655): the 40-workstream matrix `docs/testing/release-2026.08-verification.json`, the fail-closed evidence verifier `scripts/ci/verify-release-evidence.py`, the scope audit `scripts/ci/release-scope-audit.py`, strict browser mode (`scripts/run-playwright-suite.js --strict-release`) and the outcome ledger `docs/testing/release-2026.08-known-outcomes.json`. This plan does not build a parallel framework. It (1) runs every executable lane and collects baseline evidence, (2) closes the gaps the inventory exposes, (3) adds the tests and browser checks that are missing, (4) converts the eleven `external` checks into executable adapters or reviewed dispositions, and (5) ends when `verify-release-evidence.py verify` exits 0 with zero BLOCKED, SKIP, known-fail or unexpected-pass outcomes.

**Tech Stack:** Java 25, Maven, JUnit 5, AssertJ, H2, MariaDB 11.8.8 (`innodb-page-size=32768`, `sql-mode=`), JaCoCo 0.8.16; Node 22, npm 10.9.9, `node --test`; Playwright with the lockfile Chromium through `scripts/lib/playwright-harness.js`; Python 3 with the hash-pinned `scripts/ci/requirements.txt`; Ubuntu 26.04 systemd container for packaged installs (`scripts/deb-docker-validation.sh`); nginx + ModSecurity front door on `:443`.

**Spec:** Section 1 of this document (the measured scope inventory) plus `docs/testing/release-2026.08-verification.md`, `docs/release-notes-2026.08.md`, `docs/ui-tests/test-only-release-issues.md`, `docs/ui-tests/harness-coverage-5456.md`, `docs/ui-tests/app-findings-log.md` and `CLAUDE.md`. Read all of them before Task 1.

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## 0. Orientation for the executing agent

**What "since main on release/2026.08" means.** `origin/main` is at `e53e6cb5` (2026.08.0-beta6 plus the #5583 installer fix). `origin/release/2026.08` is at `710fb1ac` (2026.08.0-beta7-SNAPSHOT). `main` is a strict ancestor of the release head, so the scope is exactly `origin/main..origin/release/2026.08`: 115 commits (110 first-parent), 1,924 files, +73,461 / −16,856 lines (365 added, 1,538 modified, 21 deleted files). Everything in this plan is measured against those two SHAs. Re-measure if either branch moves.

**What already exists and must be reused.**

| Asset | Path | What it gives you |
|---|---|---|
| Verification matrix (40 workstreams, 52 checks) | `docs/testing/release-2026.08-verification.json` | The reviewed definition of done. 41 checks are executable (`junit`, `node`, `python`, `browser`, `command`, `build`); 11 are `external` and always produce BLOCKED until an adapter or disposition is committed. |
| Verifier | `scripts/ci/verify-release-evidence.py` (`validate-plan`, `list`, `run`, `verify`) | Runs a check into a private evidence directory, binds it to the source commit, tree, plan hash and candidate WAR; `verify` is the only release verdict. |
| Scope audit | `scripts/ci/release-scope-audit.py` | Complete NUL-delimited diff inventory across three baselines; refuses readiness while any production path has no `pathBindings` entry. |
| Outcome ledger | `docs/testing/release-2026.08-known-outcomes.json` | 29 browser `expectedFailure` pins, 5 retired Node TODOs, 5 retired browser expectations. Each needs a fresh installed disposition. |
| Browser manifest | `scripts/playwright-suite.json` (1,014 checks) | Tiers: core 872, extended 110, smoke 13, front-door 23, standalone 5, live-external 4; 16 manual; 7 property profiles applied by the runner itself plus the `legacy-teleplan` install profile; 29 `expectedFailure`. |
| Packaged-install harness | `scripts/deb-docker-validation.sh` (`up`, `upgrade`, `suite`, `audit`, `profile`, `shell`, `down`) | Installs the `.deb`s into a disposable Ubuntu 26.04 systemd container, stages fixtures, performs first-login reset, runs the suite behind nginx + ModSecurity. |
| Server-log audit | `scripts/deb-server-log-audit.sh`, `scripts/fixtures/server-log-signatures-2026.08.txt`, `scripts/lib/server-log-baseline.tsv` | The silent-failure detector for the installed application: any unexplained ERROR/WARN signature fails the run. |
| Changed-line coverage | `scripts/coverage/changed_line_audit.py` | Intersects a Git diff with JaCoCo XML; `--per-file`, `--fail-under`. |
| JSP directive gate | `scripts/check-jsp-directives.py` plus the `jspc` Maven profile | Static proof that the 691-JSP directive sweep (#5681) left no dead or missing directive. Static only; see Task 21 for the rendered proof. |
| Findings log | `docs/ui-tests/app-findings-log.md` (last finding: 348) | Every product defect found by a check goes here, not into the check. |

**Rules that bind every task.**

1. Topic branches start from `origin/release/2026.08` and PRs target `release/2026.08` (never `main` or `develop`). Commits are Conventional Commits with DCO sign-off (`git commit -s`). Use `test(...)` for test-only work.
2. This is test-only work. A check that exposes a product defect fails, and the defect is recorded in `app-findings-log.md` with a reproduction, then filed as a GitHub issue. Never encode a defect into a check as the expected result, never skip or quarantine a test to go green, never add a synthetic "fault endpoint" to the shipped WAR.
3. `.claude/settings.json` denies writes to `.github/**` and `database/**`. Workflow changes are written as a proposal for a maintainer in `docs/ui-tests/playwright-ci-proposal.md` (Task 28); migrations are never touched.
4. Every new Java test follows `should<Action>_<context><Condition>` (exactly one underscore), carries `@Tag("unit")` or `@Tag("integration")` plus a layer tag, extends `CarlosUnitTestBase` or `CarlosTestBase` (or is an isolated-JPA integration test with `IsolatedJpaTestDatabase`), and tests only methods that exist. Every new Node test is `scripts/<name>.test.js` run by `node --test`. Every new browser check is `scripts/<name>-playwright-checks.js` on the shared harness, owns FAKE-marked fixtures, cleans up by key, has a manifest entry and an npm alias, and is validated live on the packaged install before its PR opens.

5. SKIP is unavailable coverage, never a pass. `--strict-release` turns SKIP, known-fail, unexpected-pass, cleanup failure, missing structured result and changed deployment identity into failures. Use it for every release run.
6. Disposable targets only: `CARLOS_DISPOSABLE_HOST=true`, `ALLOW_NON_LOCAL_BASE_URL` only for a throwaway deployment, synthetic data only, no PHI anywhere in evidence, receipts or logs.
7. Evidence is immutable: commit before collecting receipts; a source or plan change invalidates the run; a failed attempt is never erased by a later retry.

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# 1. Scope inventory (measured 2026-11-06)

## 1.1 Change volume by layer

| Layer | Files in delta | Notes |
|---|---|---|
| Production Java (`src/main/java`) | 291 classes changed: 49 added, 4 deleted (`dbExtract`, `DBHandler`, `DBPreparedHandler`, `VelocityUtils`) | Heaviest packages: `commn/dao` 34, `encounter/pageUtil` 24, `billings/ca` 18, `webserv/rest` 13, `demographic/cdsimport` 10, `commn/model` 10, `report/*` 20, `utility/*` 20 |
| Web views (`src/main/webapp`, `src/main/resources`) | 717 view files: 691 `.jsp`, 20 `.jspf`, 5 `.js`, 1 `.css` | 561 files changed only in directives or comments (#5681); 31 files have 8 or more substantive changed lines (list in 1.5) |
| Routes and config | `struts-clinical.xml`, `struts-demographic.xml`, `struts-provider.xml`, `struts-scheduling.xml`, `web.xml`, `applicationContextREST.xml`, `carlos.properties`, 5 `oscarResources_*.properties`, 14 new `wsdl/published/*.wsdl`, 9 `.jrxml` deleted (BC CSV and DxResearch now rendered without Jasper) | `web.xml`: `FrmCustomedPDFServlet` replaced by `PrescriptionPdfServlet`. `carlos.properties`: `db_driver=org.mariadb.jdbc.Driver`, `db_uri=jdbc:mariadb://localhost/`, `AbandonOldChart=true` |
| Dependencies (`pom.xml`) | MySQL Connector/J 9.6.0 removed, MariaDB Connector/J 3.5.11 added; Metro, standalone Xerces, Xalan, Velocity, Ant, HttpClient 4, Jasypt, ZXing javase, `jasperreports-pdf`, `jasperreports-excel-poi`, CXF aggregate removed; `jasperreports-openpdf`, `poi-ooxml`, scoped CXF modules added; Flyway 13.12.0, Drools 10.2.0, OWASP Encoder 1.6.0, JavaMelody 2.9.0; OpenPDF 3.0.3 is vendored under `local_repo/com/github/librepdf/openpdf/` (jar, pom and parent pom); `dependencies-lock.json` and `package-lock.json` changed with these | Every removal is a compatibility question (workstream T21); the vendored jar needs a provenance disposition in Task 27 |
| Database | 0 migration files in `main..release` (the six main-only migrations V1.0.59.27 to .32 are now on both branches; the scope audit reports 0 unresolved) | Nothing to backport; upgrade path still needs the installed proof (Task 23) |
| Debian packaging | 26 files: postinst, templates, rules, lintian overrides, `carlos-emr-mariadb-init`, `carlos-emr-cert`, Flyway runner JDBC, DrugRef lock and JDBC check, new `test_backup_restic.py`, `test_modern_chart_cutover.py` | Driver cutover reaches `debian/assets/tomcat/Catalina/localhost/carlos.xml` |
| CI (`.github/workflows`) | 18 modified, 1 added (`release-verification.yml`) | Not writable from this plan; see Task 28 |
| Java tests | 2,406 → 2,490 test classes (+84 net; 262 files touched) | Includes the new native MariaDB transaction families and JDBC compatibility suites |
| Node tests | 442 → 503 `scripts/*.test.js` | `npm run test:scripts` runs all of them; `scripts/run-node-release-tests.js` is the strict release runner |
| Browser checks | manifest 937 → 1,014 entries (+78 new, 1 removed `allergy-partial-start-date`, 63 entries changed); 790+ `*-playwright-checks.js` scripts | 29 `expectedFailure` pins remain |

## 1.2 Commits that landed after the matrix was reviewed

The matrix's `reviewedRelease` is `c5350045c7`; its own commit is `8e70d7a85a` (#5655). Thirteen first-parent commits landed after the matrix and have no workstream text written for them. Each must be bound to a workstream and to named tests in Task 5, and the rows marked "add" need new coverage in the tasks named.

| Commit | Change | Risk | Owning workstream | Coverage status |
|---|---|---|---|---|
| `710fb1ac` | JDBC runtime and packaging moved to MariaDB Connector/J (`MariaDbCompatibilityDriver`, `MariaDbJdbcCompatibility`, `MariaDbPreferredTlsPlugin`, `OscarTrackingBasicDataSource`, `carlos.properties`, `src/main/webapp/META-INF/context.xml`, the packaged Tomcat `carlos.xml`, the devcontainer `volumes/carlos.properties` and `drugref2.properties`, Flyway runner, DrugRef) | Highest. Every SQL path, TLS, timeouts, metadata, upgrade of existing installs whose `carlos.properties` still names `com.mysql.cj.jdbc.Driver` | T04, T21, T26 | 7 new unit suites plus `JdbcConnectorMariaDbIntegrationTest`; no packaged upgrade proof: **add** (Tasks 12, 23) |
| `9ee16eaa` #5602 | Modern chart forced (`AbandonOldChart=true`), legacy editor writes rejected | Installs that still carry the old flag; legacy encounter editor callers | T13 | `CarlosPropertiesModernChart*UnitTest`, `test_modern_chart_cutover.py`; `legacy-encounter-editor-isolation` is a manual check: **add** installed proof (Task 23) |
| `1650895f` #5526 | Metro removed; JAXB XML adapters and `RestXmlAnnotationIntrospector` added to preserve REST XML contracts | REST XML clients, SOAP WSDL byte contracts | T08, T20, T21 | the seven `utility/xml` unit tests, `RestJaxbContextUnitTest`, `PublishedWsdlContractUnitTest` and the `rest-xml` browser check; four of the six new adapters (`CharacterIntegerMapXmlAdapter`, `StringMapXmlAdapter`, `DateXmlAdapter`, `XmlCompatibilityAdapter`) and the introspector have no direct test: **add** (Task 7) |
| `57054b73` #5553, `a257dce4` #5550 | Fixed report queries and initial legacy queries migrated to typed DAOs; `DBHandler`/`DBPreparedHandler` deleted | Report rows, totals, ordering, null handling vs. raw SQL | T34, T17 | `report-typed-queries` check, `ReportConfigurationDao`, `BilledProviderReportDao`, `AntenatalReportDao`, `DayScheduleNamesDao` integration tests; `ReportEvaluator`, `RunClinicalReport2Action` untested: **add** (Task 9) |
| `80c022cd` #5681 | Dead JSP directives removed from 691 pages, quality findings resolved | A directive that was live only through a dynamic include or EL; compiles but fails at render | T40 | `check-jsp-directives.py` plus jspc (static only): **add** rendered walk (Task 21) |
| `c5add8d9` #5560 | Report template row, output and deadline budgets | Legitimate large reports now truncated or timed out | T33, T17 | `ReportResultBudgetUnitTest`, `ReportQueryDeadlineUnitTest`, `SQLReporterCsvUnitTest`; no installed budget probe: **add** (Task 25 uses `scripts/report-budget-probe/`) |
| `7bac5ad1` #5650 | Add and edit demographic text field limits aligned | Silent truncation or refusal on edit of existing long values | T34 | `DemographicFieldLengthUnitTest`, `boundary-demographic-text` check: run on install (Task 18) |
| `3bd9b5ec` #5682 | Untyped stream length published at acquisition (`DeferredContentLength`) | Binary downloads through nginx: wrong `Content-Length`, truncated PDF/CSV | T22 | `DeferredContentLengthUnitTest`, `ResponseSanitizationFilterTomcatIntegrationTest`; `direct-response-contract` checks on install (Task 18) |
| `1e9fd6a9` #5680 | Localized date patterns and schedule fallback diagnostics | fr/es/pl/pt dates, parse round trips | T30 | `OscarResourcesBundleParseTest`; `i18n-locale-walk` extended check (Task 18) |
| `6ea14c74` #5468 | JDOM2 hardened (no external entities) | XML inputs in lab/CDS paths | T20 | `Jdom2SecurityUnitTest` |
| `dececl9b` #5495 | Nested-forward captures and response framing preserved in filters | Login, logout broadcast, CSRF script injection, first-render pages | T22 | 9 test files touched; run `login`, `logout-broadcast`, `csrf-*` checks on install (Task 18) |
| `d55d0205` #5458 | Documentation only | none | none | none |

The 19 commits between `c5350045c7` and the matrix commit (facility chooser #5615, REST pagination 400 #5596, Rich Text Letter autofill #5630, eForm POST-only editor #5651, job types DataTables #5646, null note locks #5611, BC CSV #5670, Teleplan HTML #5669, email resolutions #5649, startup key docs #5673, legacy REST appointment updates #5676, unused runtime libraries #5511, consult attachment query reuse #5677, plus five test-only commits) are already named in the matrix's `implementationBase` and are covered by its workstreams; Task 5 binds their paths.

### 1.3 Behaviour contracts that changed (breaking-change candidates)

Each row is a deliberate change whose old behaviour some caller may rely on. Each needs one named test that pins the new contract, one that proves the documented compatibility path, and a decision recorded in the `http-client-compatibility` disposition (Task 27).

| Contract | Old | New | Where pinned today |
|---|---|---|---|
| REST appointment update | last-writer-wins | `editVersion` optional; a stale token returns HTTP 409; omitted/blank keeps old behaviour (#5663, #5676) | `ScheduleServiceEndpointTest`, `AppointmentUpdateDaoIntegrationTest`, `AppointmentConverterUnitTest` |
| Startup encryption key | generated key written to the Tomcat user's home | key persisted to the loaded external configuration; classpath-only config without a key refuses to start (#5621, #5673) | `StartupKeyPersistenceUnitTest`, `StartupUnitTest`; installed restart is `external` |
| JDBC driver and URI | `com.mysql.cj.jdbc.Driver`, `jdbc:mysql:///` | `org.mariadb.jdbc.Driver`, `jdbc:mariadb://localhost/`; compatibility driver accepts the old names | `MariaDbCompatibilityDriverUnitTest`, `MariaDbJdbcCompatibilityUnitTest`, `JdbcConnectorMariaDbIntegrationTest` |
| Chart mode | `AbandonOldChart` configurable | always modern; legacy editor writes refused | `CarlosPropertiesModernChart*UnitTest` |
| REST list pagination | invalid values coerced | HTTP 400 (#5596) | `DemographicServiceEndpointTest`, `PharmacyServiceEndpointTest` |
| Patient photo, chart patient, tickler attachments, flowsheet template, quick-list read | 500 or silent | 400 / 400 / 400 / 404 / 403 (#5591, #5524, #5539, #5557, #5595) | `ClientImage2ActionTest`, `EditTickler2ActionUnitTest`, `DxResearchLoadQuickList2ActionUnitTest`, `error-sanitization` and `get-reject-*` checks |
| eForm template editor save | GET accepted | POST required (#5651) | `HtmlEdit2ActionUnitTest`, `get-reject-eform-editor` |
| BC billing pages | reachable on any install | refused on non-BC installs (#5544) | `BcBillingProvinceGuardUnitTest`, `billing-bc-province-guard` |
| BC CSV, Dx Research XLSX | Jasper-rendered | generated without Jasper; byte baselines under `src/test/resources/export-baseline/` (#5521) | `SpreadsheetExportsUnitTest`, `BcCsvDownloadUnitTest` |
| CSV cells | raw | spreadsheet formulas neutralised (#5499) | `SpreadsheetCsvUnitTest`, `SQLReporterCsvUnitTest` |
| Prescription PDF servlet | `FrmCustomedPDFServlet` | `PrescriptionPdfServlet`, half-letter printing restored (#5619) | `PrescriptionPdfServletUnitTest`; browser `rx-print-*` |
| REST XML | Jackson JakartaXmlBind introspector | `RestXmlAnnotationIntrospector` plus six adapters | `RestJaxbContextUnitTest`, `Jaxb*UnitTest`, `rest-xml` |
| Removed libraries | present on the classpath | absent (T21 list) | `LegacyDependencyCompatibilityUnitTest`, `Jdk*CompatibilityUnitTest`, `RetainedCommonsSemanticsUnitTest`, `dependency-security.test.js` |
| Deleted internal classes | `DBHandler`, `DBPreparedHandler`, `VelocityUtils`, `billings/MSP/dbExtract` | gone; `FrmCustomedPDFServlet` remains but is no longer mapped in `web.xml` | compile-time only, no external caller can reach them; `LegacyDependencyCompatibilityUnitTest`, `jsp-compile` |
| Resource bundles | | 23 keys added to `oscarResources_en.properties`, none removed; the other four locales must carry the same keys or pages render `???key???` | `OscarResourcesBundleParseTest`, `scripts/check-i18n-properties.sh`, `i18n-locale-walk` |

## 1.4 Gaps the inventory exposes

**(a) Production Java classes in the delta with no test that names them** (25: 24 after resolving `*Impl` classes to their interfaces and dropping `DemographicExtKey`, whose only change is a removed `@NotNull` annotation, plus `CheckBillingData`, whose two existing tests were not touched). The six `EctDisplay*2Action` edits are one- or two-line i18n changes and are covered as a group in Task 10.

| Area | Classes | Task |
|---|---|---|
| JAXB/REST XML | `CharacterIntegerMapXmlAdapter`, `StringMapXmlAdapter`, `DateXmlAdapter`, `XmlCompatibilityAdapter`, `RestXmlAnnotationIntrospector`, `FilterDefinitionTransfer`, `AppointmentArchiveTransfer` | 7 |
| Fax | `FaxPollBackoff` (new, 75 lines) | 8 |
| BC billing | `BcBillingRequestGuards` (new), `CheckBillingData` (100 lines changed, its two tests untouched) | 8 |
| Reports | `ReportEvaluator`, `RunClinicalReport2Action` | 9 |
| Lab | `CommonLabTestValues`, `LabVersionComparator`, `LabDisplayHelper` | 9 |
| EBS | `AttachmentCleanupInterceptor`, `DownloadInInterceptor` | 8 |
| Demographic/encounter | `DemographicSearchResultTransformer`, `EctDisplayDecisionSupportAlerts2Action`, `EctDisplayMeasurements2Action`, `EctDisplayDx2Action`, `EctDisplayDxAction`, `EctDisplayEconsult2Action`, `EctDisplayIssues2Action`, `EctDisplayResolvedIssues2Action` | 10 |

**(b) Changed browser-harness libraries without a direct Node test** (11): `scripts/lib/authz-read-fixture.js`, `authz-read-probe.js`, `concurrency-support.js`, `double-submit-helpers.js`, `hrm-report-list.js` (not referenced by any test), `mutation-families.js`, `owned-annotation-fixture.js`, `pmm-attach-source-context-checks.js`, `renal-input-fixture.js`, `throwaway-login-fixture.js`, `uncertain-server-writes.js`. Task 14.

**(c) Scope ownership.** `release-scope-audit.py --inventory-only` reports 1,085 production paths in the `currentMain` scope with no `pathBindings` entry (711 JSP/JSPF, 299 Java, 14 WSDL, 12 Debian, 8 resources, the rest config), 2,448 across all three baselines, and `ready: false`. The plan has no `pathBindings` object at all today. Task 5.

**(d) Known outcomes.** 29 browser `expectedFailure` pins (findings 118, 123, 141, 148, 153, 163, 322, 323, 330, 331, 342, 343, 344 and others), 5 retired browser expectations that still need a fresh installed PASS (findings 122, 199, 217, 261, 265), 5 retired Node TODOs (finding 166 ×4, finding 172). Finding 334 (a committed prescription whose response is lost is reported as NOT saved) is open and blocks T05. Task 22.

**(e) External checks** (11, always BLOCKED): `coverage-ratchet`, `scope-and-main-disposition`, `installed-service-restart`, `http-client-compatibility`, `ebs-packaged-staging`, `cds-native-large`, `deb-clean-upgrade-reinstall`, `backup-retention-purge`, `image-overwrite-contract`, `harness-crash-qualification`, `installed-load-budgets`. Task 27.

**(f) Java discovery.** The matrix notes two CDS classes whose `@BeforeAll` assumptions produce zero tests and zero reported skips in H2; they count as missing until the native lane executes them. Task 13.

**(g) Browser runtime budget.** Worst-case manifest timeouts sum to 79.6 h for `core`, 12.3 h for `extended`, 2.0 h for `front-door`, 1.1 h for `smoke`. The matrix requires `packaged-full` three iterations per province (ON and BC), eight property-profile runs per province, and 50 repetitions of `messenger-group-admin`. Real durations are far below the timeout sums but have never been measured for the whole tier; Task 4 measures them, and Task 17 shards across disposable containers (one container per host because of host networking).

## 1.5 View files with substantive changes (8 or more non-directive lines)

All of these already have at least one browser script that reaches their route, except where noted. They are the first targets of the rendered walk in Task 21 and of the per-route assertions in Task 18.

`report/reportonbilledphcp.jsp` (521 lines, `report-typed-queries`), `admin/jobTypes.jsp` (171, `admin-jobs`), `js/newCaseManagementView.js.jsp` (142, 28 scripts), `web/dashboard/display/drilldownDisplayController.js` (107, `dashboard-display`), `rx/RenalDosing.jsp` (51, `rx-renal-*`), `documentManager/attachDocument.jsp` (50, 9 scripts), `WEB-INF/eform-assets/editControl2.js` (43, `eform-rtl-*`), `report/reportonbilledvisitprovider.jsp` (42, `report-typed-queries`), `email/emailComposeError.jsp` (40, `eform-email-error`), `tickler/ticklerEditAttachmentsRefused.jsp` (38, `tickler-attachments`), `admin/configureFax.jsp` (27, `fax-configure`), `oscarConsultationRequest/ConsultationFormRequest.jsp` and `attachmentsNotIncludedNotice.jspf` (27 each, `consultation-*`), `rx/SearchDrug3.jsp` (27, 29 scripts), `rx/ViewScript2.jsp` (27, 18 scripts),

`admin/securityupdatesecurity.jsp` (26, `security-password-length`), `eform/attachEform.jsp` (25), `messenger/generatePreviewPDF.jsp` (20, `messenger-pdf-attachments-server-render`), `billing/CA/BC/billingDigSearch.jsp` (19), `casemgmt/uploadimage.jsp` (19, `patient-photo-upload`), `lab/CA/ALL/labDisplayAjax.jsp` (17), `admin/securityupdate.jsp` (12), `jspf/demographic-field-length-limits.jspf` (12, `boundary-demographic-text`), `tickler/ticklerDemoMain.jsp` (11), `adminFlowsheet/EditFlowsheet.jsp` (10), `rx/Preview2.jsp` (9), `admin/adminbackupdownload.jsp` (8, `direct-response-contract`), `casemgmt/newEncounterHeader.jsp` (8), `casemgmt/ticklers.jsp` (8), `demographic/add-form-personal.jsp` (8), `tickler/ticklerEdit.jsp` (8).

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## Global Constraints

- Target branch `release/2026.08`; version metadata stays `2026.08.0-beta7-SNAPSHOT`, SCM tag `HEAD`. Never edit a Flyway migration, never touch `main` or `develop`.
- Qualification runtimes exactly as the matrix states: Java 25, Node 22, npm 10.9.9, MariaDB 11.8.8 with `--innodb-page-size=32768 --sql-mode= --innodb-strict-mode=OFF`, locales `en_CA` and `fr_CA`, time zones `UTC`, `America/Toronto`, `Asia/Tokyo`, the Playwright lockfile Chromium (record the launched version in every receipt).
- Java tests: BDD names with exactly one underscore; `@Tag` hierarchy; `CarlosUnitTestBase` / `CarlosTestBase` / `IsolatedJpaTestDatabase`; `hibernateTemplate.flush()` for `HibernateDaoSupport` DAOs; boolean comparisons in HQL; no instance-identity assertions across contexts; `LogCapture` for Log4j assertions.
- Browser checks: FAKE-marked owned fixtures, cleanup by key, `assertsDatabase` and `mutates` declared in the manifest, `expectedFailure` only with a findings-log row that is not `fixed`, `timeoutSec` within the tier ceiling enforced by `playwright-suite-manifest.test.js`, `runLast` for checks that lock the shared account.
- No PHI, credentials, HTTP bodies, stdout or configuration in receipts; raw logs stay in the private run directory.
- Coverage target: every changed executable production line in `origin/main..origin/release/2026.08` is either hit by a passing named test in the H2 or native lane, or listed in a reviewed exclusion with a reason (package metadata, JRXML, generated code). Refusal branches in security and data-integrity classes (filters, guards, DAO transactions, merge, Rx stash, BCAR save, queue IDs) need 100% branch coverage. Per-file floors are ratcheted from the Task 3 baseline, never set globally.
- The 2026-10-08 coverage-gap plan (`docs/superpowers/plans/2026-10-08-playwright-coverage-gaps.md`) and the #5456 task/evidence map remain the authority for which browser workflows exist; do not duplicate a check that already exists under another name (see the crosswalk in `docs/ui-tests/coverage-publication-2026-10-18.md`).

## Review Focus

Five inputs the spec implies but no existing test exercises end to end, most likely to bite first. Each line names the task that adds its test.

1. **An existing install upgraded to the candidate with `db_driver=com.mysql.cj.jdbc.Driver` and `db_uri=jdbc:mysql:///` still in `/etc/carlos-emr/carlos.properties`, and a DrugRef `drugref2.properties` with the old JDBC defaults.** Expected: the compatibility driver accepts both, Flyway runs, DrugRef connects, `carlos-ctl check` is clean. Task 23 (`deb-upgrade-verify.sh` plus canaries) and Task 12.
2. **A binary response (prescription PDF, BC CSV, Dx Research XLSX, backup download) served through nginx after a filter reset or a nested forward.** Expected: exact `Content-Length`, no HTML appended, `%PDF` or CSV header bytes, no `CARLOS Error: 0`. Task 18 (the `direct-response-contract` check in strict mode) and Task 25.
3. **A page whose JSP lost a directive in #5681 but whose EL or scriptlet still uses that import through a dynamic `jsp:include`.** Expected: every changed view renders without 500, `???key???`, `pageerror` or a new server-log signature. Task 21.
4. **A report template that legitimately exceeds the new row, output or deadline budget.** Expected: a truthful refusal message, no partial attachment presented as complete, no silent truncation of an export the user downloads. Task 25.
5. **An install that still has `AbandonOldChart=false` (or the key absent) and a bookmarked legacy encounter editor URL.** Expected: the modern chart opens, the legacy write is refused with a status and message, no 500, no lost note. Task 23 and the `legacy-encounter-editor-isolation` manual check run by name.

---

# Phase 0: Environment and baseline evidence

## Task 1: Prepare the qualification environment

**Files:**

- Create: nothing in the repository; a fresh clone of `release/2026.08` in a disposable Ubuntu 26.04 host with Docker, JDK 25, Maven, Node 22, npm 10.9.9, MariaDB 11.8.8 client, `libxml2-utils`, `poppler-utils`, restic, nginx/ModSecurity/CRS packages available for the WAF real-engine check.

- [ ] **Step 1: Fresh clone and branch**

```
git clone https://github.com/carlos-emr/carlos.git carlos-release && cd carlos-release
git fetch origin release/2026.08 main
git checkout -b test/release-2026.08-verification origin/release/2026.08
git rev-parse HEAD   # expected 710fb1ac7c... or newer; record it
```

- [ ] **Step 2: Toolchain pins**

```
java -version 2>&1 | head -1          # 25
node --version                         # v22.x
npm install --global npm@10.9.9 --ignore-scripts --no-audit --no-fund && npm --version   # 10.9.9
npm ci --ignore-scripts --no-audit --no-fund
npx --no-install playwright install --with-deps chromium
python3 -m venv /tmp/carlos-release-verification-venv
/tmp/carlos-release-verification-venv/bin/pip install --require-hashes -r scripts/ci/requirements.txt
export PYTHON=/tmp/carlos-release-verification-venv/bin/python
```

- [ ] **Step 3: Validate the plan file and list checks**

```
"$PYTHON" scripts/ci/verify-release-evidence.py validate-plan
"$PYTHON" scripts/ci/verify-release-evidence.py list
```

Expected: `validate-plan` exits 0; `list` prints 52 check ids (41 executable, 11 `external`).

- [ ] **Step 4: Physical MariaDB with the packaged invariant**

```
docker run -d --name carlos-ci-db -p 3306:3306 -e MARIADB_ALLOW_EMPTY_ROOT_PASSWORD=1 mariadb:11.8.8 \
  --innodb-page-size=32768 --sql-mode= --innodb-strict-mode=OFF
docker exec carlos-ci-db mariadb -uroot -N -e 'SELECT @@innodb_page_size'   # 32768
```

- [ ] **Step 5: Record the environment** in `docs/testing/release-2026.08-evidence-log.md` (new file, one table: date, host, SHA, tool versions, Chromium version). This file is the running ledger every later task appends to.

## Task 2: Static gates on the release head

- [ ] **Step 1: Run every static check the PR workflows run, locally**

```
        make install                                  # in the devcontainer; elsewhere: mvn -B -ntp -DskipTests
package
        bash scripts/validate_struts_actions.sh
        bash scripts/lint/check-encoder-null-safety.sh
        bash scripts/lint/check-struts-dtd.sh
        bash scripts/lint/check-bdd-test-naming.sh
        ./scripts/check-i18n-properties.sh --output /tmp/i18n-parity-report.txt
        bash scripts/audit-i18n-coverage.sh
        mvn -B -ntp package -DskipTests -Pjspc
        mvn -B -ntp dependency:build-classpath -DincludeScope=compile -Dmdep.outputFile=target/jsp-classpath.txt
        python3 scripts/check-jsp-directives.py --classpath-file target/jsp-classpath.txt
        python3 scripts/check-war-container-apis.py target/*.war
        python3 -m unittest discover -s scripts/ci -p 'test_*.py' -v
        python3 -m unittest discover -s scripts/coverage -p 'test_*.py' -v
        python3 -m unittest discover -s scripts -p test_sonar_reliability_report.py
        python3 scripts/test_convert_jrxml_v6_to_v7.py
        python3 scripts/test-provider-signature-migration.py
        python3 scripts/test-wcb-reference-migration.py
        PROV=on bash scripts/ci/verify-flyway-baseline.sh    # Task 1 MariaDB plus the Flyway CLI at pom.xml's
flyway.version; repeat with PROV=bc
        pip install bcrypt==5.0.0 && python3 -m unittest discover -s debian/assets/tests
        python3 -m unittest discover -s scripts/migration/o19/tests -t .
        python3 scripts/waf/check-clinical-policy.py --check-config
        python3 scripts/waf/check-clinical-policy.py --check-inventory
```

Expected: every command exits 0. Record each exit code in the evidence log. A non-zero exit here is a product or packaging defect: file it before continuing.

- [ ] **Step 2: Pinned carlos-ctl suite against this tree** (what `debian-python-tests.yml` runs; the CLI is its own repository, pinned by `debian/carlos-ctl.pin`):

```
        tag="$(sed -n 's/^tag=//p' debian/carlos-ctl.pin)"
        git clone --quiet --depth 1 --branch "$tag" https://github.com/carlos-emr/carlos-ctl.git /tmp/carlos-ctl
# or the pin's fallback commit
        python3 scripts/ci/prepare-cli-manifest-test-copy.py /tmp/carlos-ctl /tmp/carlos-ctl-current-data "$PWD"
        ( cd /tmp/carlos-ctl-current-data && CARLOS_SRC="$OLDPWD" CARLOS_CTL_SRC="$PWD" PYTHONPATH="$PWD"
PYTHONDONTWRITEBYTECODE=1 python3 -m unittest discover -s tests -t . )
        python3 scripts/ci/prepare-cli-manifest-test-copy.py --verify /tmp/carlos-ctl /tmp/carlos-ctl-current-data
"$PWD"
```

- [ ] **Step 3: Static analysis the PR workflows run.** PMD with `.github/pmd/carlos-ruleset.xml` (`pmd check --dir src/main/java --dir src/test/java --rulesets .github/pmd/carlos-ruleset.xml --format sarif --report-file /tmp/pmd.sarif`, the PMD binary at the version `pmd.yml` pins) and SpotBugs (`mvn -B compile spotbugs:spotbugs`, then `python3 scripts/ci/spotbugs-xml-to-sarif.py`). Run the same commands on a `main` checkout and diff the SARIF: every new finding in a changed file is triaged into a product fix or the narrowest suppression `docs/static-analysis-workflows.md` allows, never ignored.

- [ ] **Step 4: Scope inventory**

```
        "$PYTHON" scripts/ci/release-scope-audit.py --inventory-only --output /tmp/scope-baseline.json
        python3 -c "import json;s=json.load(open('/tmp/scope-baseline.json'));print({k:s[k] for k in
('changedPathsAcrossScopes','unownedProductionPaths','unresolvedMainOnlyMigrations','ready')})"
```

Expected today: `changedPathsAcrossScopes 8987, unownedProductionPaths 2448, unresolvedMainOnlyMigrations 0, ready False`. Keep the file; Task 5 drives it to `unownedProductionPaths 0`.

## Task 3: Baseline Java evidence (H2 lane, time zones, coverage)

- [ ] **Step 1: Run the authoritative H2 lane through the verifier**

```
        "$PYTHON" scripts/ci/verify-release-evidence.py run --check java-all --evidence /tmp/carlos-release-
evidence
```

Expected: receipt outcome `PASS`; `{run}/coverage/jacoco.xml` present. If any test fails, that is a release blocker: record it and file it before any coverage work.

- [ ] **Step 2: Time-zone lane**

```
        for tz in America/Toronto Asia/Tokyo; do
          "$PYTHON" scripts/ci/verify-release-evidence.py run --check java-date-zones --profile "$tz" --evidence
/tmp/carlos-release-evidence
        done
```

- [ ] **Step 3: Changed-line coverage baseline**

```
        RUN=<the runId the java-all run printed>
        python3 scripts/coverage/changed_line_audit.py /tmp/carlos-release-evidence/$RUN/coverage/jacoco.xml
origin/main HEAD --per-file > /tmp/coverage-baseline.txt
```

Expected output: one line per changed production file with covered/missed counts and missed line numbers, and the overall changed-line percentage. Paste the summary into the evidence log. The matrix's earlier audit against the older workspace main measured 82.4% of changed executable lines; your number against `origin/main` is the ratchet floor for Task 11.

- [ ] **Step 4: Discovery reconciliation**

```
        "$PYTHON" scripts/ci/verify-java-discovery.py --source src/test/java --reports /tmp/carlos-release-
evidence/$RUN/reports --output /tmp/java-discovery-baseline.json
```

Expected: exit 1 with the two CDS families with zero executed cases reported as `missing` in the JSON output; nothing else is missing. Record the family names for Task 13.

## Task 4: Baseline browser evidence and tier timing

- [ ] **Step 1: Build the candidate and the packages**

```
        export CARLOS_RELEASE_DEB_VERSION="$(dpkg-parsechangelog -SVersion)"
        "$PYTHON" scripts/ci/verify-release-evidence.py run --check build --evidence /tmp/carlos-release-evidence
        # Then build the three .debs following docs/ui-tests/deb-install-validation.md sections 1-3,
        # passing the candidate WAR as CARLOS_WAR so the package carries the same bytes.
```

- [ ] **Step 2: Bring up the Ontario install and run the smoke tier**

```
        CARLOS_DISPOSABLE_HOST=true DEBS_DIR=/path/to/debs scripts/deb-docker-validation.sh up
        CARLOS_DISPOSABLE_HOST=true scripts/deb-docker-validation.sh suite --tier smoke --strict-release --
residue-audit
        # the wrapper adds its own --junit and copies the report to LOG_DIR; do not pass --junit twice
        CARLOS_DISPOSABLE_HOST=true scripts/deb-docker-validation.sh audit
```

Expected: 12/12 smoke checks PASS, `residue audit: no residue`, server-log audit reports no unexplained signature. The thirteenth smoke entry, `login-smoke-no-pin`, carries `profile: no-pin` and runs only inside the no-pin profile run (Task 19). The last recorded packaged smoke result was 12/12 on Chromium 154.0.8025.0 (`harness-coverage-5456.md`).

- [ ] **Step 3: Measure the core and extended tiers once, diagnostic mode, to learn their real duration**

```
        CARLOS_DISPOSABLE_HOST=true scripts/deb-docker-validation.sh suite --tier core --province ON
        CARLOS_DISPOSABLE_HOST=true scripts/deb-docker-validation.sh suite --tier extended --province ON
```

Record wall-clock per check from the `time` attributes of the JUnit file the wrapper copies to `LOG_DIR`. Use the totals to size the shards in Task 17. Every failure in this run is classified, not fixed: known-fail (listed in the ledger), product defect (findings log), harness defect (`test-only-release-issues.md`).

- [ ] **Step 4: Commit the evidence log**

```
        git add docs/testing/release-2026.08-evidence-log.md
        git commit -s -m "test(release): record 2026.08 qualification baseline"
```

---

# Phase 1: Traceability

## Task 5: Bind every changed production path to a test, check and workstream

**Files:**

- Modify: `docs/testing/release-2026.08-verification.json` (add a top-level `pathBindings` object)
- Create: `scripts/ci/generate-path-bindings.py`, `scripts/ci/test_generate_path_bindings.py`

**Interfaces:**

- Produces: `pathBindings[path] = {"reason": str, "tests": [existing files], "checks": [check ids], "workstreams": [T##]}` exactly as `release-scope-audit.py` validates (every field non-empty, every test path an existing file, every check and workstream known).

- [ ] **Step 1: Write the failing test** `scripts/ci/test_generate_path_bindings.py`:
  - `test_should_emit_binding_for_java_file_with_matching_test` (a `Foo.java` with `FooUnitTest.java` present → binding with that test, check `java-all`, workstream from the package map).
  - `test_should_leave_unmatched_java_file_unbound` (no candidate → path absent from output, listed in the "unbound" report).
  - `test_should_bind_jsp_to_browser_check_by_route` (a JSP named in a `struts-*.xml` result whose action name appears in a `*-playwright-checks.js` → binding with that script and `packaged-full`).
  - `test_should_reject_binding_naming_missing_test`.
- [ ] **Step 2: Run** `python3 -m unittest scripts/ci/test_generate_path_bindings.py -v` → FAIL (module missing).
- [ ] **Step 3: Implement** `generate_bindings(root: Path, base: str, head: str, package_workstreams: dict[str, str]) -> tuple[dict, list[str]]` in `scripts/ci/generate-path-bindings.py`. Candidate rules, in order: a Java file binds to any `src/test/**/<Stem>(Unit|Integration|Endpoint)?Test.java`, to tests that `import` or name the class, and to the interface's tests for `*Impl`; a JSP/JSPF binds to the browser scripts that name its Struts route (parse `<action name>` and `<result>` from `struts-*.xml`) and to `jsp-compile`; `package-info.java`, `.jrxml`, `.wsdl`, `.properties`, Debian and `pom.xml` entries bind to the matrix check that already covers them (`PublishedWsdlContractUnitTest`, `JasperTemplateFontPolicyUnitTest`, `OscarResourcesBundleParseTest`, `debian-python`, `java-legacy-compat`) with a reason string. Print the unbound list to stderr; never invent a test.
- [ ] **Step 4: Run** the unit test → PASS. Then run the generator and merge its output into the plan:

```
        python3 scripts/ci/generate-path-bindings.py --base origin/main --head HEAD --plan docs/testing/release-
2026.08-verification.json --write
        "$PYTHON" scripts/ci/verify-release-evidence.py validate-plan
        "$PYTHON" scripts/ci/release-scope-audit.py --inventory-only --output /tmp/scope-after.json
```

Expected: `validate-plan` still exits 0; `unownedProductionPaths` drops from 2,448 to the generator's unbound count (expect roughly the 28 classes of section 1.4a plus the 561 directive-only JSPs until Task 21 adds the rendered walk check they bind to).

- [ ] **Step 5: Hand-write the remaining bindings** for the 25 untested classes (they point at the tests Tasks 7 to 10 create; create the bindings in those tasks' commits, not here) and for every path whose correct owner the generator could not infer. Each hand-written reason names the behaviour, not the file.
- [ ] **Step 6: Commit**

```
        git add scripts/ci/generate-path-bindings.py scripts/ci/test_generate_path_bindings.py
docs/testing/release-2026.08-verification.json
        git commit -s -m "test(release): bind changed production paths to executable evidence"
```

## Task 6: Workstream text for the thirteen post-matrix commits

**Files:**

- Modify: `docs/testing/release-2026.08-verification.json` (extend `sources`, `cases` and `acceptance` of T04, T08, T13, T17, T20, T21, T22, T26, T30, T33, T34, T40), `docs/testing/release-2026.08-verification.md` (one bullet per commit under "Implemented regressions and gates")

- [ ] **Step 1:** For each row of section 1.2, add the commit's production files to the owning workstream's `sources`, add its failure cases to `cases`, and add the test or check that pins it to `requiredChecks` where a new check id is introduced later in this plan (`jsp-render-walk`, `report-budget-probe`; register the ids in `checks` now with `kind: external` and the reason "adapter added by Task N", so the plan validates). Also register two checks the matrix lacks although CI runs them: `o19-generator-python` (`python3 scripts/ci/run-python-release-tests.py --start scripts/migration/o19/tests --output {run}/python.json`, required by T25 and T26) and `carlos-ctl-pinned-suite` (`kind: command`, Task 2 Step 2 wrapped in `scripts/ci/run-pinned-carlos-ctl-suite.sh`, required by T26); bind `scripts/migration/o19/**` and `debian/assets/o19-manifest/o19map_schema.json` to them.
- [ ] **Step 2:** `validate-plan` → exit 0. `python3 scripts/ci/run-python-release-tests.py --start scripts/ci --output /tmp/py.json` → all pass.
- [ ] **Step 3: Commit** `test(release): own post-matrix commits in the verification matrix.`

---

# Phase 2: Java unit coverage gaps

## Task 7: JAXB adapters and REST XML introspector

**Files:**

- Create: `src/test/java/io/github/carlos_emr/carlos/utility/xml/CharacterIntegerMapXmlAdapterUnitTest.java`, `StringMapXmlAdapterUnitTest.java`, `DateXmlAdapterUnitTest.java`, `XmlCompatibilityAdapterUnitTest.java`; `src/test/java/io/github/carlos_emr/carlos/webserv/rest/util/RestXmlAnnotationIntrospectorUnitTest.java`; `src/test/java/io/github/carlos_emr/carlos/webserv/rest/to/model/FilterDefinitionTransferXmlUnitTest.java`; `src/test/java/io/github/carlos_emr/carlos/webserv/transfer_objects/AppointmentArchiveTransferXmlUnitTest.java`
- Reference: existing `CalendarXmlAdapterUnitTest`, `CharacterXmlAdapterUnitTest`, `JaxbUnmarshalCompatibilityUnitTest`, baselines under `src/test/resources/xml/`

- [ ] **Step 1: Write the failing tests.** Per adapter: `shouldMarshalEmptyMap_toEmptyElement`, `shouldRoundTripMap_withUnicodeKeys`, `shouldUnmarshalNull_toNull`, `shouldRejectDuplicateKeys_withIllegalArgument` (or document the keep-last rule if that is what the Metro-era baseline did: compare against `jaxb-moxy-baseline.json`). For `DateXmlAdapter`: `shouldMarshalDate_inUtcIsoFormat`, `shouldUnmarshalLegacyMetroFormat_toSameInstant`, `shouldFailClosed_onGarbage`. For the introspector: `shouldHonorXmlElementName_overJsonProperty`, `shouldIgnoreTransientFields_forXml`, `shouldMatchMetroBaseline_forAppointmentTo1` (marshal `AppointmentTo1` and compare canonicalised XML to `src/test/resources/xml/jaxb-character-array-baseline.xml` style fixtures; add a new baseline captured from the `main` build if none exists for that type, and say in the test where it came from).
- [ ] **Step 2:** `mvn -B -ntp test -Dtest='*XmlAdapterUnitTest,RestXmlAnnotationIntrospectorUnitTest,*TransferXmlUnitTest' -Dtest.forkCount=1` → FAIL (compilation).
- [ ] **Step 3: Implement nothing in production.** If a test cannot pass without a production change, the adapter is wrong: record a finding and stop this task.
- [ ] **Step 4:** Re-run → PASS. Add `pathBindings` rows for the seven classes.
- [ ] **Step 5: Commit** `test(xml): pin JAXB adapter and REST XML introspector contracts.`

## Task 8: Fax backoff, BC request guards, EBS interceptors, CheckBillingData

**Files:**

- Create: `src/test/java/io/github/carlos_emr/carlos/fax/core/FaxPollBackoffUnitTest.java`, `src/test/java/io/github/carlos_emr/carlos/billings/ca/bc/pageUtil/BcBillingRequestGuardsUnitTest.java`, `src/test/java/io/github/carlos_emr/carlos/integration/ebs/client/ng/AttachmentCleanupInterceptorUnitTest.java`, `DownloadInInterceptorUnitTest.java`
- Modify: `src/test/java/io/github/carlos_emr/carlos/billings/MSP/CheckBillingDataUnitTest.java` (create if the two existing references are not a dedicated test)

- [ ] **Step 1: Write the failing tests.** `FaxPollBackoff`: `shouldStartAtBaseDelay_afterFirstFailure`, `shouldDoubleUntilCap_onConsecutiveFailures`, `shouldResetToZero_afterSuccess`, `shouldNotStarveHealthyAccount_whenAnotherFails`, `shouldReportRetryAt_forStatusPage` (read the exact base, multiplier and cap from the class; put those numbers in the assertions). `BcBillingRequestGuards`: `shouldRefuse_whenInstallIsNotBc` (asserts the SecurityException message uses the paren form),

`shouldAllow_whenBillingRegionIsBc`, `shouldRefuseGet_forMutatingRoute`. Interceptors: `shouldDeleteTemporaryAttachment_afterMessageHandled`, `shouldNotDeleteOnFault_whenCleanupDisabled`, `shouldBoundDownloadToConfiguredLimit`. `CheckBillingData`: one test per changed branch shown by `git diff origin/main..HEAD -- src/main/java/io/github/carlos_emr/carlos/billings/MSP/CheckBillingData.java` (100 lines), named for the validation rule each branch implements.

- [ ] **Step 2:** Run with `-Dtest=` → FAIL. **Step 3:** no production edits. **Step 4:** PASS; bindings. **Step 5: Commit** `test(billing,fax,ebs): cover untested release classes`.

## Task 9: Report evaluation, clinical report action, lab helpers

**Files:**

- Create: `src/test/java/io/github/carlos_emr/carlos/report/ClinicalReports/ReportEvaluatorUnitTest.java`, `src/test/java/io/github/carlos_emr/carlos/report/ClinicalReports/PageUtil/RunClinicalReport2ActionUnitTest.java`, `src/test/java/io/github/carlos_emr/carlos/lab/ca/on/CommonLabTestValuesUnitTest.java`, `src/test/java/io/github/carlos_emr/carlos/lab/ca/all/util/LabVersionComparatorUnitTest.java`, `src/test/java/io/github/carlos_emr/carlos/lab/ca/all/web/LabDisplayHelperUnitTest.java`

- [ ] **Step 1: Failing tests.** `ReportEvaluator`: the typed-DAO path versus the retired raw SQL for numerator/denominator (`shouldComputeSameRatio_asLegacyQueryFixture` using a small H2 fixture through `IsolatedJpaTestDatabase`), `shouldRefuseTemplate_whenDeadlineExceeded`, `shouldNotLogClinicalPayload_onFailure` (LogCapture). `RunClinicalReport2Action`: `shouldThrowSecurityException_withoutReportPrivilege`, `shouldReturnNone_afterStreamingCsv`, `shouldSendError_whenGenerationFailsBeforeWrite`. `CommonLabTestValues`: the #5608 provenance change (`shouldLinkMeasurement_toSourceReportId`, `shouldRemainDeterministic_withDuplicateResults`). `LabVersionComparator`: ordering of equal, newer and malformed versions. `LabDisplayHelper`: the one changed line, named for its behaviour.
- [ ] **Steps 2 to 5** as in Task 7. Commit `test(report,lab): cover typed report evaluation and lab helpers`.

## Task 10: Encounter panel actions and demographic transformer

**Files:**

- Modify: `src/test/java/io/github/carlos_emr/carlos/encounter/pageUtil/EncounterPanelPrivilegeRegressionUnitTest.java` (add a parameterised case per `EctDisplay*2Action` in section 1.4a asserting the i18n key or privilege object it now uses)
- Create: `src/test/java/io/github/carlos_emr/carlos/commn/DemographicSearchResultTransformerUnitTest.java`

- [ ] **Step 1:** The transformer test pins the two changed lines (#5501 name bulk-load) with a `DemographicName` projection fixture. `DemographicExtKey` changed only by dropping a `@NotNull` annotation and its import: it needs no test and is bound in Task 5 as a reviewed non-executable change.
- [ ] **Steps 2 to 5** as above. Commit `test(encounter,demographic): pin panel privileges and name projection`.

## Task 11: Changed-line coverage ratchet and the `coverage-ratchet` adapter

**Files:**

- Create: `scripts/coverage/release-coverage-floors.json` (per-file floors from the Task 3 baseline, raised by Tasks 7 to 10), `scripts/coverage/check-coverage-floors.py`, `scripts/coverage/test_check_coverage_floors.py`
- Modify: `docs/testing/release-2026.08-verification.json` (`coverage-ratchet` becomes `kind: command` running the new script against the java-all run's `jacoco.xml`)

- [ ] **Step 1: Failing test:** `test_should_fail_when_file_drops_below_floor`, `test_should_fail_on_unmapped_executable_source`, `test_should_require_full_branch_coverage_for_security_classes` (the list: `HttpMethodGuardFilter`, `ResponseSanitizationFilter`, `LogoutBroadcastFilter`, `CsrfGuardScriptInjectionFilter`, `LoginFilter`, `BcBillingRequestGuards`, `DemographicMergedDaoImpl`, `BCAR2020FormSaveService`, `RxWriteScript2Action` stash path, `QueueDaoImpl`, `MariaDbCompatibilityDriver`, `DeferredContentLength`).
- [ ] **Step 2 to 4:** implement `check_floors(jacoco_xml: Path, floors: Path, base: str, head: str) -> int`, reusing `changed_line_audit.read_jacoco_lines` and `changed_lines`; run `changed_line_audit.py ... --fail-under <baseline>` and the new script; both exit 0.

- [ ] **Step 5:** Re-run `java-all` through the verifier, then `coverage-ratchet`. Expected: PASS. Commit `test(coverage): ratchet changed-line floors for release 2026.08`.

---

# Phase 3: Java integration and native MariaDB

## Task 12: Native lanes with Connector/J, both schemas, both provinces

- [ ] **Step 1:** Run the three native checks through the verifier against the Task 1 server (URL must end in `/`, no credentials in it; credentials in the native-test environment variables):

```
export CARLOS_TEST_MARIADB_DAO=true CARLOS_TEST_MARIADB_URL=jdbc:mariadb://127.0.0.1:3306/
for c in java-native-generated java-native-flyway java-native-bc-flyway; do
  "$PYTHON" scripts/ci/verify-release-evidence.py run --check "$c" --evidence /tmp/carlos-release-evidence
done
```

Expected: PASS for all three with zero skips; `JdbcConnectorMariaDbIntegrationTest` executes in the generated lane; the BCAR, Rx rollback, consultation projection and patient merge families execute in both Flyway lanes.

- [ ] **Step 2: Add the native proofs the matrix's Review Focus needs but no test has:** in `src/test/java/io/github/carlos_emr/carlos/utility/JdbcConnectorMariaDbIntegrationTest.java` add `shouldConnectWithLegacyMysqlDriverName_throughCompatibilityDriver` (datasource configured with `com.mysql.cj.jdbc.Driver` and `jdbc:mysql:///carlos_test`), `shouldApplyConfiguredTimeouts_onMariaDbUrl` (uses `CdsJdbcConnectionTimeouts`), `shouldNegotiateTls_withPreferredPlugin_whenServerOffersIt` (uses `src/test/resources/jdbc/compatibility-server-cert.pem`), `shouldReportSameMetadataDefaults_asMysqlConnector` (against `JdbcMetadataDefaultUnitTest` fixtures).
- [ ] **Step 3:** Run the suite alone (`-Dtest=JdbcConnectorMariaDbIntegrationTest -Dgroups=mariadb-native`) → PASS; re-run `java-native-generated` through the verifier → PASS.
- [ ] **Step 4: Commit** `test(jdbc): prove Connector/J cutover against physical MariaDB`.

## Task 13: Discovery gaps: the two CDS families and parameterised templates

- [ ] **Step 1:** For each family Task 3 Step 4 reported, read its `@BeforeAll` assumption. If it requires the native lane, add the class name to the `-Dtest=` list of `java-native-flyway` in the plan JSON and to `.github/workflows/maven-project.yml` **proposal** (Task 28). If it can run in H2 with a fixture, add the fixture.
- [ ] **Step 2:** Re-run `verify-java-discovery.py` → no missing family. Commit `test(cds): execute every discovered test family in a required lane`.

---

# Phase 4: Node and Python

## Task 14: Direct tests for the eleven untested harness libraries

**Files:**

- Create: `scripts/authz-read-fixture.test.js`, `scripts/authz-read-probe.test.js`, `scripts/concurrency-support.test.js`, `scripts/double-submit-helpers.test.js`, `scripts/hrm-report-list.test.js`, `scripts/mutation-families.test.js`, `scripts/owned-annotation-fixture.test.js`, `scripts/pmm-attach-source-context-checks.test.js`, `scripts/renal-input-fixture.test.js`, `scripts/throwaway-login-fixture.test.js`, `scripts/uncertain-server-writes.test.js`

- [ ] **Step 1:** For each library, write `node --test` cases against its exported functions with a fake `page`/`sql` runner (follow `scripts/owned-installer.test.js` and `scripts/property-profiles.test.js` for the pattern). Minimum per library: one happy path, one refusal path, one cleanup-on-failure path. For `uncertain-server-writes.js` specifically: `a lost response after dispatch marks the write pending and retains the fixture`; `a failure before dispatch does not mark pending`. For `throwaway-login-fixture.js`: `the fixture login is deleted by key after the check, even on SIGTERM` (use `scripts/lib/script-signal-safety.js`).

- [ ] **Step 2:** `node --test scripts/<name>.test.js` → FAIL (missing). **Step 3:** no library changes unless a test exposes a harness bug (then fix the library and say so in the commit). **Step 4:** `npm run test:scripts` → all pass; `node scripts/run-node-release-tests.js --output /tmp/node.json` → `Node release tests: all cases passed.`
- [ ] **Step 5: Commit** `test(harness): cover release 2026.08 harness libraries directly`.

### Task 15: Strict Node and Python evidence

- [ ] **Step 1:**

```
        "$PYTHON" scripts/ci/verify-release-evidence.py run --check node-all --evidence /tmp/carlos-release-
evidence
        "$PYTHON" scripts/ci/verify-release-evidence.py run --check evidence-self-tests --evidence /tmp/carlos-
release-evidence
        "$PYTHON" scripts/ci/verify-release-evidence.py run --check coverage-self-tests --evidence /tmp/carlos-
release-evidence
        "$PYTHON" scripts/ci/verify-release-evidence.py run --check waf-self-tests --evidence /tmp/carlos-release-
evidence
        "$PYTHON" scripts/ci/verify-release-evidence.py run --check debian-python --evidence /tmp/carlos-release-
evidence
        for f in fixture-measurement-whtr fixture-rtl-measurement fixture-eform-toolbar fixture-dashboard-tickler-
picker fixture-calendar-eform fixture-prevention-print-popup fixture-echart-note-lifecycle fixture-hrm-report-list
fixture-messenger-reply-form fixture-search-sort-messenger-inbox fixture-clinical-risk sql-hook waf-config waf-
inventory; do
          "$PYTHON" scripts/ci/verify-release-evidence.py run --check "$f" --evidence /tmp/carlos-release-evidence
        done
```

Expected: every receipt PASS. `node-all` fails on any TODO or SKIP: the five retired TODOs (finding 166 ×4, 172) must be active passing regressions, not `todo`.

- [ ] **Step 2: The remaining host-side checks**, each with its named prerequisite (a missing variable makes the verifier record BLOCKED with "Missing named test prerequisites", never a pass):

```
        "$PYTHON" scripts/ci/verify-release-evidence.py run --check jsp-compile --evidence /tmp/carlos-release-
evidence
        "$PYTHON" scripts/ci/verify-release-evidence.py run --check scope-inventory --evidence /tmp/carlos-
release-evidence
        "$PYTHON" scripts/ci/verify-release-evidence.py run --check waf-real-engine --evidence /tmp/carlos-
release-evidence   # nginx, ModSecurity and CRS installed
        CARLOS_SMOKE_CHROMIUM=/path/to/chromium CARLOS_SMOKE_CHROMEDRIVER=/path/to/chromedriver \
          "$PYTHON" scripts/ci/verify-release-evidence.py run --check java-browser-pdf --evidence /tmp/carlos-
release-evidence
        LEGACY_COMPATIBILITY_REPO=/path/to/legacy/maven/repository \
          "$PYTHON" scripts/ci/verify-release-evidence.py run --check java-legacy-compat --evidence /tmp/carlos-
release-evidence
        CARLOS_CTL_SRC=/tmp/carlos-ctl \
          "$PYTHON" scripts/ci/verify-release-evidence.py run --check java-development-seed --evidence
/tmp/carlos-release-evidence
```

Read `LegacyDependencyCompatibilityUnitTest` for what the legacy repository must contain, and `EFormBrowserPdfServiceSeleniumSmokeIntegrationTest` for the Chromium and chromedriver pair it expects.

---

## Phase 5: Playwright against the packaged install

### Task 16: Classify every result of the Task 4 diagnostic core run

**Files:**

- Modify: `docs/ui-tests/app-findings-log.md` (new rows from 349), `docs/ui-tests/test-only-release-issues.md`, `scripts/playwright-suite.json` (only to retire a pin whose finding is now `fixed`, never to add one for a new defect without a findings row)

- [ ] **Step 1:** For every non-PASS result: reproduce it by hand on the same install; decide product defect (findings log row with reproduction, GitHub issue, `expectedFailure` pin naming the exact step) or harness defect (fix the script, add the Node regression, record in `test-only-release-issues.md`).
- [ ] **Step 2:** Re-run only the affected checks with `--only` in strict mode → each is PASS or known-fail at exactly its pinned step.

- [ ] **Step 3: Commit** `test(playwright): classify release 2026.08 core-tier outcomes`.

## Task 17: `packaged-full`, three iterations per province

- [ ] **Step 1: Where and how packaged checks run.** Every `packaged: true` check runs **inside** the validation container, never from the host: the verifier attests that each file under `/usr/share/carlos-emr/webapp/carlos` matches the candidate WAR before and after the command, and the runner needs the loopback front door and the MariaDB socket. `scripts/deb-docker-validation.sh up` mounts this checkout read-only at `/root/carlos` and writes `BASE_URL`, `MYSQL_*`, `CHROME_PATH` and `EXPECT_FRONT_DOOR=true` into `/root/suite-env.sh`. Inside `scripts/deb-docker-validation.sh shell`: create the hash-pinned venv under `/root`, copy the candidate WAR to `/root/candidate/carlos.war`, keep evidence under `/root/evidence` (outside the read-only source), `source /root/suite-env.sh`, then run the verifier from `/root/carlos`. Copy each run directory out to the host's evidence root when it ends; never edit a receipt.

- [ ] **Step 2: Iterations are whole runs.** The verifier executes each check's fixed command as one process and `--strict-release` fails on an empty or partial result set, so one iteration is one unsharded full run of the province's selection. Parallelise across iterations and provinces, not inside one: one validation container per host (host networking), so three ON and three BC containers on six hosts, or fewer hosts in sequence, sized from the Task 4 timings. Only if an iteration cannot finish inside one host's window, extend `scripts/ci/release_verification.py` with shard-aware receipts (a `shards` field on the check; `verify` accepts an iteration only when the union of the shard selections equals the unsharded selection and every shard passed), with the regression in `scripts/ci/test_release_adapters.py` that a missing or overlapping shard is refused.

- [ ] **Step 3: Run** (inside the ON container; then the same with `--profile BC` inside the BC container):

```
. /root/suite-env.sh
for i in 1 2 3; do
  "$PYTHON" scripts/ci/verify-release-evidence.py run --check packaged-full --profile ON --iteration $i \
    --artifact /root/candidate/carlos.war --evidence /root/evidence
done
```

Expected: six PASS receipts. A failed iteration stays in the evidence (duplicate identities are refused); fix the cause, bump nothing, and run the next iteration number.

- [ ] **Step 4:** Run `scripts/deb-docker-validation.sh audit` after every iteration and attach the server-log audit summary to the evidence log. Any new signature not in `scripts/fixtures/server-log-signatures-2026.08.txt` is a finding.

## Task 18: Per-commit browser assertions for the commits with Java-only tests

Thirty-two commits in the delta changed production code and carry Java tests but touched no browser check or Node test (list in Appendix A, column P = 0 and N = 0). For each, name the existing check that exercises the changed route on the install, or add one.

- [ ] **Step 1:** Build the table in `docs/testing/release-2026.08-evidence-log.md`: commit, route(s) touched, check name(s), run id. Existing checks to use: `direct-response-contract` (#5682, #5495, #3566), `login`, `login-facility-chooser`, `logout-broadcast`, `csrf-runtime-forms`, `csrf-xhr-token` (#5495, #5615), `rest-xml` and `rest-roster-enrolled-to` and `oauth-rest-surfaces` (#5526, #5596, #5676), `consultation-*` (#5563, #5614, #5677), `billing-bc-*` (#5552, #5669, #5670), `fax-configure` and `fax-queue-admin` (#5612), `rx-*` (#5600, #5619, #5627, #5590), `echart-*` and `concurrency-chart-note-takeover` (#5629, #5593, #5611), `report-typed-queries` and `report-by-template` (#5550, #5553, #5560), `boundary-demographic-text` (#5650), `i18n-locale-walk` (#5680), `eform-*` (#5498, #5630, #5651), `application-health` and `error-sanitization` (#5621, #5673).
- [ ] **Step 2:** New checks where no route is reached today: `schedule-day-names-bulk` (#5501: a day with 40 appointments renders every patient name with one name query, assert via `assertsDatabase` on the slow log or query count endpoint if available, else on response time bounds and content), `lab-measurement-source-link` (#5608: a lab-derived measurement in the flowsheet links to its source report and refuses another patient's report id), `consult-list-no-n-plus-one` (#5563: 50 consultations list in one page load; assert row count, ordering and absence of other patients' names), `encounter-navbar-logout-race` (#5629: open the encounter, log out in a second window, interact with the navbar: no 500). Each gets a manifest entry (`extended`, `assertsDatabase` as appropriate), npm alias, FAKE fixtures, and a live PASS on the install before commit.

- [ ] **Step 3: Commit** per check: `test(playwright): <behaviour>`.

## Task 19: Property-profile runs and the 50-repetition series

The seven property profiles (`address-lock`, `appointment-options`, `caisi-admin`, `legacy-contacts`, `multisite`, `no-pin`, `private-consent`) are applied by the runner itself: `scripts/lib/property-profiles.js` appends the profile's keys to `/etc/carlos-emr/carlos.properties`, restarts `carlos-emr` through `systemctl`, runs the profile's checks in a child process, fails if the service restarted during them, and restores the file byte for byte before restarting again. Only `legacy-teleplan` (BC install, `NEW_BC_TELEPLAN=no`) is switched with `scripts/deb-docker-validation.sh profile legacy-teleplan` and switched back with `profile default`; that command refuses every other name.

- [ ] **Step 1:** Inside the container (Task 17 Step 1), serially, because two profiles cannot own the properties file at once:

```
. /root/suite-env.sh
for p in address-lock appointment-options caisi-admin legacy-contacts multisite no-pin private-consent; do
  "$PYTHON" scripts/ci/verify-release-evidence.py run --check packaged-profile-$p --profile ON \
    --artifact /root/candidate/carlos.war --evidence /root/evidence
done
# BC container: the same seven with --profile BC; then, from the host,
#   CARLOS_DISPOSABLE_HOST=true scripts/deb-docker-validation.sh profile legacy-teleplan
#   (inside) run --check packaged-profile-legacy-teleplan --profile BC --artifact ... --evidence ...
#   CARLOS_DISPOSABLE_HOST=true scripts/deb-docker-validation.sh profile default
```

Expected: PASS for each, properties file restored (the runner verifies), no service restart counted during the checks. A check that SKIPs inside its own profile run fails the run by design.

- [ ] **Step 2:** `packaged-repeat-5566`, iterations 1 to 50, ON then BC, inside the containers. Any failure is recorded with its iteration number; the series keeps its failure disposition.

## Task 20: Front-door, standalone and live-external tiers

- [ ] **Step 1:** `suite --tier front-door --strict-release` on both installs with `EXPECT_FRONT_DOOR=true`: the WAF corpus family, `eform-signature-data-uri`, `deb-login` (manual, by name), `report-query-by-example-front-door`. Expected: PASS; `waf-real-engine` through the verifier → PASS.
- [ ] **Step 2:** `--tier standalone` on the build host (no install needed). **Step 3:** `--tier live-external` and the four `fax-*` manual scripts under `scripts/e2e/fax/` only with authorised SRFax test credentials; otherwise record them as BLOCKED with the reason, never as PASS.

## Task 21: Rendered walk of every changed view (the JSP directive sweep proof)

**Files:**

- Create: `scripts/generate-changed-view-routes.py`, `scripts/test_generate_changed_view_routes.py`, `scripts/fixtures/release-2026.08-changed-views.json`, `scripts/jsp-render-walk-playwright-checks.js`, `scripts/jsp-render-walk.test.js`
- Modify: `scripts/playwright-suite.json` (entry `jsp-render-walk`, tier `core` and `extended`, `provinces: ["all"]`, `assertsDatabase: false`, `timeoutSec` from the measured run), `package.json` (`test:jsp-render-walk-playwright`), `docs/testing/release-2026.08-verification.json` (check `jsp-render-walk`, `kind: browser`, required by T40)

**Interfaces:**

- Produces: `release-2026.08-changed-views.json` = `[{ "view": "WEB-INF/jsp/...", "routes": ["admin/ViewJobTypes"], "method": "GET", "gate": "_admin" | null, "includedBy": [...] }]` generated from `git diff --name-only origin/main..HEAD -- src/main/webapp` joined with `<action>`/`<result>` pairs in `struts-*.xml` and with static `<%@ include %>` / `<jsp:include>` graphs. Views reachable only through POST results (`attachmentsRefused`, `photoError`, `eFormError`) are listed with the mutating route and the parameter that drives the error result, so the walk can reach them with a deliberate refusal.

- [ ] **Step 1: Failing Python test:** `test_should_map_jsp_to_its_action_names`, `test_should_follow_static_include_to_parent_routes`, `test_should_mark_post_only_results`, `test_should_fail_on_view_with_no_route` (a changed view nobody can reach is dead code: a finding, not a skip).

- [ ] **Step 2:** Implement, run, commit the generated fixture (it is reviewed data: 717 views, expect roughly 600 distinct routes).
- [ ] **Step 3: Failing Node test for the check's assertions** (`jsp-render-walk.test.js`): given a fake page that returns 500, `CARLOS Error`, a `???oscar.key???` literal, a `pageerror`, a console error, or a body under 200 bytes for a known-rich page, the walker reports exactly that route and continues; given a 403 on a gated route with the clinician login, the walker reports `refused-as-expected` only when the fixture says `gate`.
- [ ] **Step 4:** Implement the check on the harness and the page-health engine (`scripts/lib/page-health-engine.js`: the same opener and `assertHealthy` that `i18n-locale-walk` and the `surface-audit` family use), with the route list taken from the fixture rather than catalogued from the UI. `anonymous-access-refused` deliberately walks only what the UI offers; the directive sweep touched pages no one-level crawl reaches. Log in as the clinician, open an owned FAKE patient, GET every route (with the patient context parameters the harness's `provider-navigation-fixture` supplies), assert per route: status in {200, 302-to-login-only-for-logout-routes, 403-only-when-gated}, no `CARLOS Error`, no `???`, no `Exception/at io.github` text, no `pageerror`, no `console.error`, `Content-Type` consistent with the view; POST-result views are reached through their documented refusal and asserted the same way. Report one line per route; the server-log window before and after is diffed with `deb-server-log-audit.sh`.
- [ ] **Step 5:** Run on the ON and BC installs in strict mode. Expected: 0 failing routes. Every failing route is a finding (log it) and the check keeps failing until the product is fixed.
- [ ] **Step 6:** Bind the 561 directive-only JSPs to this check in `pathBindings` (regenerate with Task 5's generator; add the rule "a JSP in the changed-views fixture binds to `jsp-render-walk`"). Scope audit → `unownedProductionPaths` 0.
- [ ] **Step 7: Commit** `test(playwright): render every view changed by the 2026.08 directive sweep`.

## Task 22: Retake and disposition of every known outcome

- [ ] **Step 1:** For each of the 29 `expectedFailure` pins and 5 retired expectations, run the check by name in strict mode on the current install. Outcomes: `known-fail` at the pinned step → leave the pin, confirm the issue is open and linked; PASS → remove the pin, mark the finding `fixed` with the run id; `failed-elsewhere` → a new finding.
- [ ] **Step 2:** Finding 334: run `rx-save-failure-retry-*-lost` by name; it remains a known-fail until a product fix lands on `release/2026.08`. The release cannot pass `verify` while T05 names it; record that explicitly in the evidence log rather than relaxing T05.
- [ ] **Step 3:** Update `release-2026.08-known-outcomes.json` dispositions with run ids. `validate-plan` → 0. Commit `test(release): disposition known outcomes against the candidate`.

## Task 23: Upgrade path, driver cutover and modern-chart cutover on the installed package

**Files:**

- Modify: `scripts/deb-upgrade-canaries.js`, `scripts/deb-upgrade-snapshot.test.js` (two new canaries), `docs/testing/release-2026.08-verification.json` (`deb-clean-upgrade-reinstall` becomes `kind: command` running the wrapper below)
- Create: `scripts/deb-clean-upgrade-reinstall-check.sh`, `scripts/deb-clean-upgrade-reinstall-check.test.js`

`scripts/deb-docker-validation.sh upgrade <tag>` already does most of this: it downloads the published package set for `<tag>` with `scripts/lib/deb-release-assets.py` (GitHub digests verified), installs it fresh with demo data, copies the candidate `.deb`s in, runs `scripts/deb-upgrade-canaries.js` across the upgrade (seed canaries, upgrade, `deb-upgrade-verify.sh`), then the smoke tier and the read-only core checks, and fails if `deb-upgrade-baseline.sh` differs afterwards.

- [ ] **Step 1: Failing Node test** for two new canaries in `deb-upgrade-canaries.js`: (a) the pre-upgrade `/etc/carlos-emr/carlos.properties` carries the pre-cutover `db_driver=com.mysql.cj.jdbc.Driver` and `db_uri=jdbc:mysql:///`, and DrugRef's `drugref2.properties` the old JDBC defaults; after the upgrade the application starts, `carlos-ctl check` is clean and a seeded FAKE patient is readable; (b) the pre-upgrade file carries `AbandonOldChart=false`; after the upgrade the chart opens modern and a legacy encounter editor write is refused with a status and a message, not a 500.
- [ ] **Step 2:** Implement the canaries, then run:

```
        CARLOS_DISPOSABLE_HOST=true DEBS_DIR=/path/to/candidate/debs scripts/deb-docker-validation.sh upgrade
2026.08.0-beta6
```

Expected: canaries, verifier, smoke and read-only core pass; a seeded encrypted field decrypts after the upgrade (startup key preserved); `carlos-ctl check` clean; no new server-log signature.

- [ ] **Step 3: Reinstall leg.** `scripts/deb-clean-upgrade-reinstall-check.sh` runs `upgrade 2026.08.0-beta6`, then reinstalls the identical candidate packages inside the container (`dpkg -i` of the same files), re-runs `deb-upgrade-verify.sh`, asserts that Flyway applied zero migrations on the reinstall and that database, configuration and keys are byte-identical, then runs `legacy-encounter-editor-isolation` by name (a manual check) in strict mode. The Node test pins the script's pass/fail decision against captured outputs.
- [ ] **Step 4:** Run `deb-clean-upgrade-reinstall` through the verifier; PASS. Commit `test(deb): prove upgrade and reinstall from 2026.08.0-beta6 across the JDBC and chart cutovers`.

---

# Phase 6: Breaking-change and silent-failure sweeps

## Task 24: Client compatibility matrix (the `http-client-compatibility` adapter)

**Files:**

- Create: `scripts/rest-client-compatibility-playwright-checks.js`, `scripts/rest-client-compatibility.test.js`, `scripts/fixtures/rest-client-compatibility-cases.json`
- Modify: manifest entry `rest-client-compatibility` (core), plan check `http-client-compatibility` → `kind: browser`

- [ ] **Step 1:** Cases file: one entry per row of section 1.3 that an HTTP client sees: JSON and XML `Accept` for demographic, schedule, pharmacy, consultation list; `POST /schedule/getAppointment` without `editVersion`, with a current one, with a stale one (expect 200, 200, 409); invalid pagination (400); the XML payload shape against the Metro-era baselines; OAuth 1.0a signed requests through `oauth-rest-surfaces` helpers; CSV/XLSX downloads compared byte-for-byte to `src/test/resources/export-baseline/` after normalising dates.
- [ ] **Step 2:** Node test proves the case loader refuses an entry without an expected status and content type. **Step 3:** check runs on the install, strict. **Step 4:** Any intentional break (there are two documented: `editVersion` conflict detection, startup key) is recorded in the plan check's `disposition` with the release-notes anchor. Commit `test(rest): pin supported client contracts for 2026.08`.

## Task 25: Report budgets, binary responses and export truthfulness

**Files:**

- Create: `scripts/report-budget-probe-playwright-checks.js` (drives `scripts/report-budget-probe/ReportTemplateBudgetProbe.java` fixtures already in the tree), `scripts/report-budget-probe.test.js`
- Modify: manifest (`report-budget-probe`, extended, `assertsDatabase: true`), plan (`report-budget-probe` check required by T33 and T17)

- [ ] **Step 1:** Seed a template that returns rows just under, at, and over the row budget, one whose output exceeds the output budget, and one that sleeps past the deadline (through the probe fixture, never through production code). Assert: under → complete CSV with exact row count; at → complete; over → refusal text names the budget and no partial file is offered; deadline → refusal, no `CARLOS Error: 0`, no half-written attachment in the messenger or email path (#5606 error page shown instead).
- [ ] **Step 2:** the `direct-response-contract` check in strict mode plus a new `binary-response-framing` case: prescription PDF, BC CSV, Dx Research XLSX, backup download through nginx: `Content-Length` equals body bytes, magic bytes correct, no trailing HTML. Commit `test(report): prove budgets and binary framing on the install`.

## Task 26: Silent-failure sweep

- [ ] **Step 1:** After every full iteration: `deb-server-log-audit.sh` (unexplained signatures), `--residue-audit` output, `application-health`, `error-sanitization`, `phi-in-error-pages`, `audit-log-*` family. All must be PASS with no new signature.

- [ ] **Step 2:** The harness already records every `console` message and `pageerror` per check and reports them as problems (`scripts/lib/playwright-harness.js`), and `scripts/lib/page-health-engine.js` already judges page text (a literal `null`, `undefined`, `NaN` or `Invalid Date`, an unresolved `???key???`, a leaked exception name, the CARLOS error page) for the audit checks. Prove with a Node test whether a check built on `runWorkflow` fails when a navigated document carries those text signals; if it does not, route the engine's text judgment through the harness for every navigated document behind an explicit per-check opt-out, with the test covering both directions. Re-run smoke. Commit `test(harness): judge page text on every navigated document`.
- [ ] **Step 3:** `installed-load-budgets` adapter: `scripts/installed-load-budget-check.js` samples JVM heap, open connections (`OscarTrackingBasicDataSource` counters through JavaMelody's authenticated endpoint) and thread counts before and after the smoke tier; fails when the after value does not return within the documented bound. Plan check → `kind: command`.

---

# Phase 7: External checks, CI proposal, final verdict

## Task 27: Adapters or dispositions for the remaining external checks

For each, either a committed executable adapter (preferred) or a reviewed disposition object in the plan with the reason it cannot be executed here and who owns it:

| Check | Plan |
|---|---|
| `coverage-ratchet` | Task 11 adapter |
| `scope-and-main-disposition` | Task 5 bindings; main-only migrations now 0; `kind: command` running `release-scope-audit.py` without `--inventory-only` |
| `installed-service-restart` | `scripts/installed-service-restart-check.sh`: restart `carlos-emr` three times under load, assert the startup key is reused (ciphertext round trip of a seeded encrypted field), no ephemeral key, no partial configuration; Node/Python test for the assertion logic |
| `http-client-compatibility` | Task 24 |
| `deb-clean-upgrade-reinstall` | Task 23 |
| `installed-load-budgets` | Task 26 Step 3 |
| `harness-crash-qualification` | `scripts/signal-cleanup-audit.js` driven by a real `kill -INT` / `kill -TERM` of a running check mid-mutation on the install; assert SQL residue 0 and configured file bytes/ownership/mode restored; the historical midpoint inventory is not accepted |
| `backup-retention-purge` | `debian/assets/tests/test_backup_restic.py` already covers unit level; adapter runs the shipped `carlos-emr-backup` in the container, restores to a second container, diffs synthetic files byte for byte, checks no credential canary in any snapshot |
| `ebs-packaged-staging` | Disposition: requires Ministry test credentials; BLOCKED with owner named, never PASS |
| `cds-native-large` | Run the existing `cds-import-native.yml` and `cds-import-performance.yml` procedures (`scripts/ci/run-cds-native-tier.py`, `run-cds-performance.py`) against the candidate; adapter wraps them |
| `image-overwrite-contract` | Disposition: maintainer decision on same-name upload semantics (finding tracked); until decided, BLOCKED |

- [ ] Commit each adapter with its own test: `test(release): execute <check> as evidence`.

## Task 28: CI proposal for the maintainer

**Files:**

- Modify: `docs/ui-tests/playwright-ci-proposal.md` (append a 2026-11 section; `.github/**` is write-denied)

- [ ] **Step 1:** Propose, with exact YAML diffs: (a) `script-regressions.yml` runs the `standalone` tier (measured about 10 s); (b) a `playwright-smoke.yml` on pull requests to `release/**` that runs `--tier smoke --strict-release --province ON` against the devcontainer stack; (c) `maven-project.yml` native lanes include the Task 13 families; (d) `release-verification.yml` also runs `verify-java-discovery.py` and `check-coverage-floors.py` against the H2 lane's JaCoCo; (e) the nightly sharded core tier from the existing proposal, updated with the Task 4 timings. State what was run and what was only read.

### Task 29: Final verdict

- [ ] **Step 1:** Every check in `list` has a receipt for every required profile and iteration on the final SHA (a source change after Task 27 invalidates earlier receipts: re-run the invalidated ones, do not patch receipts). Receipts produced inside validation containers or on other hosts are copied, run directory by run directory, into the one evidence root `verify` reads (`*/receipt.json`); run directories are UUID-named so they never collide, and a copied receipt is never edited.
- [ ] **Step 2:**

```
        "$PYTHON" scripts/ci/verify-release-evidence.py verify --artifact <war> --evidence /tmp/carlos-release-
evidence --output /tmp/carlos-release-readiness.json
        echo $?
```

Expected: exit 0, `readiness.json` with zero BLOCKED, SKIP, FAIL, duplicate or stale entries, and the installed-byte binding equal to the candidate WAR. If finding 334 or any other product defect is still open, the exit is non-zero and the report says exactly which workstream blocks; that is the truthful end state, and the final message to the user lists those blockers with their issue numbers.

- [ ] **Step 3:** Write `docs/testing/release-2026.08-verification-report.md`: SHA, WAR hash, package hashes, Chromium version, per-layer counts (Java cases, Node cases, browser checks ×iterations), coverage before/after, findings opened and fixed, blockers. Open the PR(s) against `release/2026.08` with the signature block from CLAUDE.md.

---

## Self-review notes

- Spec coverage: every row of sections 1.2, 1.3 and 1.4 names the task that owns it; the eleven external checks are each assigned in Task 27; the five Review Focus items each name their task.
- Nothing in this plan edits production code, migrations or workflows. A task that cannot pass without a product change stops and records a finding.
- Type consistency: `pathBindings` shape (Task 5) is the one `release-scope-audit.py` validates; new check ids (`jsp-render-walk`, `report-budget-probe`, `rest-client-compatibility`, `o19-generator-python`, `carlos-ctl-pinned-suite`) are registered in Task 6 before the tasks that implement them.
- Proportion: this plan is shorter than the matrix it drives; commands are given where the implementer could otherwise pick a wrong lane or flag.

---

## Appendix A: Commit to test-layer map (`origin/main..origin/release/2026.08`, first-parent)

Columns: J = Java test files touched, N = Node test files touched, P = Playwright or browser-check scripts touched, M = manifest touched. A row with J > 0 and N = P = 0 is in Task 18's table.

| SHA | Subject | Prod Java | Views | J | N | P | M |
|---|---|---|---|---|---|---|---|
| `710fb1ac7c` | fix(db): migrate JDBC runtime and packaging to MariaDB Connector/J | 10 | 2 | 32 | 0 | 0 | 0 |
| `9ee16eaa0e` | fix(chart): force modern charts and reject legacy editor writes | 8 | 12 | 14 | 3 | 3 | 1 |
| `d55d02055c` | docs(security): clarify JSTL embedded Xalan scanner triage | 0 | 0 | 0 | 0 | 0 | 0 |
| `1650895f4e` | fix(deps): preserve XML contracts while removing Metro | 25 | 16 | 15 | 0 | 0 | 0 |
| `57054b737d` | refactor(report): migrate fixed report queries and bound legacy reporting SQL | 29 | 4 | 14 | 0 | 2 | 1 |
| `80c022cdb6` | fix(jsp): remove dead directives and resolve quality findings | 0 | 703 | 3 | 3 | 2 | 0 |
| `c5add8d97e` | fix(reports): enforce template row, output and deadline budgets | 4 | 0 | 5 | 0 | 1 | 0 |
| `a257dce422` | refactor(db): migrate initial legacy queries to typed DAOs | 8 | 0 | 6 | 0 | 0 | 0 |
| `7bac5ad1fa` | fix(demographic): align add and edit text field limits | 2 | 2 | 4 | 1 | 1 | 1 |
| `3bd9b5ec6f` | fix(filters): publish an untyped stream's length at acquisition | 2 | 0 | 4 | 0 | 0 | 0 |

| SHA | Subject | Prod Java | Views | J | N | P | M |
|---|---|---|---|---|---|---|---|
| `1e9fd6a9ff` | fix(i18n): correct localized date patterns and schedule fallback diagnostics | 0 | 4 | 1 | 0 | 0 | 0 |
| `6ea14c74ab` | fix(xml): enforce security safeguards for retained JDOM2 | 1 | 0 | 1 | 0 | 0 | 0 |
| `decec19b56` | fix(filters): preserve nested-forward captures and response framing | 7 | 1 | 9 | 0 | 0 | 0 |
| `8e70d7a85a` | test(release): implement verification matrix and regression gates | 6 | 1 | 10 | 5 | 2 | 0 |
| `8f3086a8b0` | chore(deps): remove unused runtime libraries | 1 | 1 | 0 | 0 | 0 | 0 |
| `e9dff829de` | fix(consult): reuse attachment queries in request view | 3 | 1 | 2 | 0 | 1 | 1 |
| `b96e367945` | Merge pull request #5676 from carlos-emr/codex/fix-5663-legacy-appointment-updates | 3 | 0 | 3 | 0 | 0 | 0 |
| `5c590826d7` | Merge pull request #5661 from carlos-emr/claude/beta-release-main-6nkj52 | 8 | 7 | 6 | 2 | 4 | 0 |
| `289091834f` | fix(startup): document classpath encryption key requirements | 1 | 0 | 3 | 0 | 0 | 0 |
| `abdba03e25` | fix(email): persist failed email resolutions and handle missing logs | 4 | 0 | 5 | 0 | 1 | 1 |
| `0928bcfc54` | fix(billing): publish Teleplan HTML only after assigning its filename | 3 | 0 | 3 | 0 | 0 | 0 |
| `a7e9862516` | fix(billing): keep BC CSV exports usable with malformed legacy rows | 3 | 0 | 3 | 0 | 1 | 0 |
| `27107fa458` | fix(casemgmt): hydrate legacy null note locks | 1 | 0 | 1 | 0 | 1 | 1 |
| `a718676ea1` | fix(admin): stop job type DataTables reinitialization on release | 0 | 6 | 1 | 0 | 2 | 1 |
| `b0699e5b67` | test(echart): validate the expected chart note takeover conflict | 0 | 0 | 0 | 1 | 1 | 1 |
| `1a159b4188` | fix(eform): require POST for template editor saves | 2 | 0 | 3 | 1 | 1 | 1 |
| `56a8700bd3` | test(notes): fix Browse Notes confirmation and fixture cleanup (#5636) | 0 | 0 | 0 | 1 | 0 | 0 |
| `d4ffdfc9fa` | test(echart): validate boundary note editing by saved identity | 0 | 0 | 0 | 0 | 2 | 1 |
| `0b0cb368b4` | test(playwright): retire four stale expected-failure flags | 0 | 0 | 0 | 0 | 0 | 1 |
| `37a48abf47` | fix(login): serve facility chooser assets before selection | 1 | 0 | 1 | 0 | 1 | 1 |
| `bac72ac0c3` | fix(rest): return 400 for invalid list pagination | 2 | 0 | 3 | 0 | 1 | 1 |
| `d86b63cce2` | fix(hooks): require SQL context before blocking Java edits | 0 | 0 | 0 | 0 | 0 | 0 |
| `b53f09d87b` | fix(eforms): stop false Rich Text Letter autofill warnings | 0 | 1 | 0 | 2 | 2 | 0 |
| `c5350045c7` | fix(startup): persist encryption keys to loaded configuration | 2 | 1 | 4 | 0 | 0 | 0 |
| `0300da57d8` | fix(rx): restore half-letter prescription PDF printing (#3331) | 0 | 2 | 0 | 3 | 1 | 1 |
| `688875683d` | fix(casemgmt): remove unused group-issue chart lookups | 3 | 0 | 2 | 1 | 1 | 1 |
| `d0fb3982d5` | fix(encounter): avoid navbar NPE during concurrent logout | 22 | 0 | 2 | 0 | 0 | 0 |
| `f54b867e26` | fix(email): show eForm preparation failures on a dedicated error page | 1 | 7 | 2 | 0 | 1 | 1 |
| `e312994e06` | fix(consult): warn about omitted attachments and retain hidden reports | 8 | 9 | 5 | 0 | 0 | 0 |
| `2b184d9f13` | fix(rx): make new prescription saves atomic (#5627) | 1 | 0 | 4 | 0 | 1 | 1 |
| `fd1ea84fd8` | fix(rest): restore XML responses on the session REST API | 3 | 0 | 1 | 0 | 1 | 1 |
| `e415c7f9a1` | test(harness): repair release 2026.08 test-only failures | 0 | 0 | 1 | 19 | 17 | 1 |
| `0581f15247` | fix(rx): finish renal validation and CI security checks | 1 | 2 | 0 | 2 | 2 | 0 |
| `04ad333bbd` | fix(privacy): omit clinical payloads from prescription diagnostics | 4 | 0 | 3 | 0 | 0 | 0 |
| `e07e92e98c` | fix(measurements): link lab-derived readings to source reports | 5 | 8 | 2 | 1 | 1 | 1 |
| `c56b019a49` | fix(filters): honor full response resets across decorators (#3566) | 4 | 0 | 1 | 0 | 0 | 0 |
| `967d117550` | fix(rx): preserve renal refresh context and refuse invalid observations | 1 | 2 | 1 | 2 | 1 | 1 |
| `61429a7fc8` | fix(fax): back off failing inbound accounts and show retry status | 2 | 6 | 1 | 0 | 0 | 0 |
| `ed248feb4a` | fix(casemgmt): join notes when looking up linked issues | 2 | 0 | 2 | 0 | 0 | 0 |
| `83ab2f3a49` | fix(casemgmt): return HTTP 400 for invalid patient photo requests | 1 | 2 | 1 | 0 | 1 | 0 |
| `2662c8683d` | fix(documents): use the created queue ID and stop on insert failure | 4 | 0 | 4 | 0 | 1 | 1 |
| `47a8aa9e6a` | test(eform): validate replay after post-save failure (#4424) | 0 | 0 | 0 | 0 | 1 | 1 |
| `9c5addc5c7` | fix(form): commit complete BCAR 2020 versions atomically | 3 | 0 | 3 | 0 | 0 | 0 |
| `68af38f12a` | fix(echart): include pending CPP saves in encounter barrier (#4733) | 0 | 2 | 1 | 3 | 1 | 0 |
| `16e696bbfb` | fix(rx): initialize custom notes with zero quantity (#5597)i | 1 | 0 | 1 | 0 | 1 | 0 |

| SHA | Subject | Prod Java | Views | J | N | P | M |
|---|---|---|---|---|---|---|---|
| 199c90c97d | fix(tests): preserve PMM attachment program policy | 0 | 0 | 0 | 2 | 1 | 1 |
| 383db89793 | fix(dashboard): reject reinterpreted tickler dates and times (#4457) (#5610) in | 0 | 1 | 0 | 1 | 2 | 0 |
| a24c566608 | fix(demographic): reject merges with missing patient records | 2 | 1 | 1 | 0 | 1 | 0 |
| 6772f3fc5c | fix(rx): prevent limited-use code link errors | 0 | 2 | 0 | 1 | 1 | 1 |
| d9cb4066a2 | test(hrm): match owned report links with patient context | 0 | 0 | 0 | 0 | 5 | 0 |
| 7de08891b1 | fix(documents): confirm Browse Notes document deletion | 0 | 6 | 0 | 1 | 1 | 1 |
| 337cb36d8f | fix(labels): make error probes deterministic and retain safe PDF diagnostics | 1 | 0 | 2 | 1 | 1 | 0 |
| 13547aeeff | fix(dxresearch): return 403 when quick-list read access is denied | 1 | 0 | 1 | 0 | 1 | 1 |
| 1cd4ae9fb9 | test(harness): report missing validation prerequisites as skips | 0 | 0 | 0 | 4 | 5 | 1 |
| eba0178bdd | test(inbox): isolate quick-pick routing provider | 0 | 0 | 0 | 1 | 1 | 1 |
| 33fd525d38 | test(sonar): guard retained legacy BC form exclusions | 0 | 0 | 0 | 1 | 0 | 0 |
| b541e0f7c0 | perf(consultations): eliminate list N+1 queries (#3890) | 19 | 1 | 7 | 0 | 0 | 0 |
| e07597273b | fix(schedule): reject stale REST appointment edits | 9 | 0 | 8 | 0 | 0 | 0 |
| 1d363415e0 | Merge pull request #5564 from carlos-emr/fix/3954-hrm-duplicates | 2 | 9 | 3 | 2 | 1 | 0 |
| adb5fde506 | fix(flowsheet): return 404 for missing editor templates | 0 | 1 | 0 | 1 | 1 | 0 |
| 12f84e6701 | refactor(billing): migrate BC MSP and WCB reads to typed DAOs | 10 | 0 | 8 | 0 | 0 | 0 |
| 6b66fc9938 | fix(labs): handle an absent primary public key quietly | 1 | 0 | 1 | 0 | 0 | 0 |
| 980a1be04b | fix(rx): reject malformed satellite clinic PDF addresses | 1 | 1 | 1 | 1 | 1 | 0 |
| 87a237180c | fix(schedule): preserve appointment count validation responses | 1 | 0 | 2 | 0 | 0 | 0 |
| 8678d11821 | fix(tickler): return HTTP 400 for refused edit attachments | 1 | 3 | 2 | 1 | 1 | 0 |
| 7b65c40ec0 | fix(messenger): preserve refused reply drafts and verify Send and Archive | 0 | 1 | 0 | 0 | 2 | 0 |
| 6633ee9fd9 | fix(billing): refuse BC-only pages on non-BC installs | 4 | 0 | 2 | 1 | 1 | 1 |
| f5c66abeff | docs: standardize library choices for new CARLOS code | 0 | 0 | 0 | 0 | 0 | 0 |
| 4ef47a2654 | fix(reports): generate BC CSV and Dx Research XLSX without Jasper | 9 | 9 | 26 | 0 | 3 | 1 |
| 0cbe677539 | fix: handle absent provider billing session preference | 1 | 0 | 1 | 0 | 0 | 0 |
| be931c3fcb | fix(debian): enforce dependency locks and repair lintian overrides | 0 | 0 | 0 | 1 | 0 | 0 |
| 3e19ccf936 | test(search): recognize WAF refusals of deliberate attack probes | 0 | 0 | 0 | 1 | 2 | 0 |
| 9eb03174d9 | fix(prevention): avoid error log storms for missing patients | 1 | 0 | 1 | 1 | 0 | 0 |
| 8298445437 | test(messenger): fix descending inbox sort validation | 0 | 0 | 0 | 0 | 2 | 1 |
| ff17d2feda | refactor(db): retire unused raw JDBC connection accessor | 2 | 0 | 6 | 1 | 0 | 0 |
| 866f2f318f | fix(deb): retire default IPv6 nginx site before first apply | 0 | 0 | 0 | 1 | 0 | 0 |
| cc91a71f70 | fix(test): match chart context in notes pagination checks | 0 | 0 | 0 | 1 | 1 | 0 |
| 13a679bd76 | fix(calendar): restore hidden keyboard calendar controls | 0 | 3 | 0 | 0 | 2 | 0 |
| 933822657d | fix(clinical): return HTTP 400 for malformed chart patients | 1 | 0 | 1 | 0 | 2 | 0 |
| ac8e3fb8b3 | fix(reports): unify JasperReports and application PDFs on OpenPDF 3 | 0 | 0 | 2 | 0 | 0 | 0 |
| 96b88aa847 | test(harness): cover release 2026.08 clinical workflows and cleanup | 0 | 0 | 0 | 76 | 215 | 1 |
| 112feaecb7 | fix(deps): remove CXF HTTP 4 with scoped EBS transport | 3 | 0 | 3 | 0 | 0 | 0 |
| ad7ce1c2ff | fix(xml): migrate CDS to JDK XML and remove standalone Xerces | 6 | 0 | 3 | 0 | 0 | 0 |
| 9748d012cd | chore(deps): replace CXF aggregate with required runtime modules | 0 | 1 | 0 | 0 | 0 | 0 |
| 7d72f44d49 | fix(deps): upgrade Flyway to 13.12.0 on release 2026.08 | 0 | 0 | 0 | 0 | 0 | 0 |
| 69f5a7f0c9 | fix(reports): neutralize spreadsheet formulas in CSV exports | 3 | 0 | 4 | 0 | 1 | 0 |
| 070929a53e | fix(deps): upgrade Drools to 10.2.0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 450c28e0d4 | fix(test): install annotation font in Debian validation image | 0 | 0 | 0 | 0 | 0 | 0 |
| 3f693289c4 | fix(eform): copy ZIP contents with bounded bulk I/O | 1 | 0 | 2 | 0 | 0 | 0 |
| 2dd9fcbbfb | refactor(strings): consolidate legacy StringUtils helpers | 9 | 1 | 6 | 0 | 0 | 0 |

| SHA | Subject | Prod Java | Views | J | N | P | M |
|---|---|---|---|---|---|---|---|
| `fe88bc1ebe` | refactor: use behavior-equivalent JDK APIs for encoding, streams and collections | 39 | 1 | 6 | 0 | 1 | 0 |
| `1c411c36ed` | refactor(deps): remove unused legacy libraries | 14 | 0 | 8 | 1 | 0 | 0 |
| `8def72af8e` | perf(schedule): bulk-load patient names in day responses | 6 | 0 | 3 | 0 | 0 | 0 |
| `dcb2111213` | chore(deps): exclude unused OpenTelemetry SDK from Selenium | 0 | 0 | 0 | 0 | 0 | 0 |
| `c31a9635e2` | test(rx): name patient in literal-null pharmacy preview | 0 | 0 | 0 | 1 | 1 | 0 |
| `7115e3eed1` | test(echart): honor shared patient-scope refusal statuses | 0 | 0 | 0 | 0 | 1 | 0 |
| `69974cb315` | test: install Liberation Sans in deb validation image | 0 | 0 | 0 | 0 | 0 | 0 |
| `23154da28f` | fix(security): upgrade OWASP Encoder to 1.6.0 | 4 | 4 | 12 | 0 | 0 | 0 |
| `7d0f7256e6` | fix(deps): upgrade JavaMelody to 2.9.0 | 0 | 0 | 2 | 0 | 0 | 0 |
| `9c66215173` | fix(security): remove Jasypt from EBS WS-Security runtime | 2 | 0 | 1 | 0 | 0 | 0 |

Subjects are shown without their PR number suffix; the SHA resolves the PR.

## Appendix B: Command reference

```
        # Verifier
        "$PYTHON" scripts/ci/verify-release-evidence.py validate-plan | list
        "$PYTHON" scripts/ci/verify-release-evidence.py run --check <id> --evidence DIR [--profile P] [--iteration
N] [--artifact WAR]
        "$PYTHON" scripts/ci/verify-release-evidence.py verify --evidence DIR --artifact WAR --output
readiness.json
        "$PYTHON" scripts/ci/release-scope-audit.py [--inventory-only] --output scope.json

        # Java
        make install --run-tests                  # all tests, H2
        mvn test -Dgroups="unit" | -Dgroups="integration"
        mvn -B -ntp test jacoco:report -Dtest.forkCount=3
        python3 scripts/coverage/changed_line_audit.py target/site/jacoco/jacoco.xml origin/main HEAD --per-file -
-fail-under <pct>
        CARLOS_TEST_MARIADB_DAO=true CARLOS_TEST_MARIADB_SCHEMA=flyway
CARLOS_TEST_MARIADB_URL=jdbc:mariadb://127.0.0.1:3306/ mvn test -Dtest=<families> '-Dgroups=spring-context |
mariadb-native'
        mvn -B -ntp package -DskipTests -Pjspc && python3 scripts/check-jsp-directives.py --classpath-file
target/jsp-classpath.txt

        # Node and Python
        npm run test:scripts                      # node --test scripts/*.test.js
        node scripts/run-node-release-tests.js --output node.json
        python3 -m unittest discover -s scripts/ci -p 'test_*.py'
        python3 -m unittest discover -s debian/assets/tests -p 'test_*.py'

        # Playwright on the packaged install
        CARLOS_DISPOSABLE_HOST=true DEBS_DIR=... scripts/deb-docker-validation.sh up
        CARLOS_DISPOSABLE_HOST=true scripts/deb-docker-validation.sh suite --tier smoke --strict-release --
residue-audit --junit out.xml
        CARLOS_DISPOSABLE_HOST=true scripts/deb-docker-validation.sh suite --only <check> --strict-release --
residue-audit
        CARLOS_DISPOSABLE_HOST=true scripts/deb-docker-validation.sh profile <name> | profile default
        CARLOS_DISPOSABLE_HOST=true scripts/deb-docker-validation.sh audit
        node scripts/run-playwright-suite.js --list
        bash scripts/deb-upgrade-baseline.sh; bash scripts/deb-upgrade-verify.sh; node scripts/deb-upgrade-
canaries.js
```